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Chapter 1

Introduction

Non-Newtonian and viscoelastic fluid flows are ubiquitous across modern industrial, chemical, and biomedical engineering, from polymer processing, extrusion, and injection molding to microfluidic lab-on-a-chip devices and physiological biofluid transport. Unlike classical Newtonian fluids, polymer solutions and macromolecular melts exhibit fading memory, non-linear stress-strain relationships, and intense elasticity. These characteristics give rise to remarkable hydrodynamic phenomena, such as rod climbing (the Weissenberg effect), die swell, and purely elastic flow instabilities occurring even in the vanishing Reynolds number regime where inertia is negligible.

Modeling and simulating these complex flows computationally represents a longstanding challenge in continuum mechanics. Conventional numerical discretization techniques—principally the Finite Element Method (FEM) and Finite Volume Method (FVM)—are severely hindered by the celebrated High Weissenberg Number Problem (HWNP), wherein hyperbolic stress advection produces exponential boundary layers and steep localized gradients near stagnation points and solid walls, frequently causing catastrophic numerical divergence. Furthermore, while standard computational fluid dynamics (CFD) packages excel at forward predictive simulations when all constitutive rheological parameters are pre-

cisely known, extracting unknown fluid properties from experimental observations (such as velocity measurements) requires computationally prohibitive outer-loop optimization workflows.

Over the past decade, Scientific Machine Learning (SciML) and Physics-Informed Neural Networks (PINNs) have emerged as a transformative computational paradigm. By embedding the governing continuous partial differential equations (PDEs) directly into neural network loss formulations via automatic differentiation (AD), PINNs seamlessly bridge forward continuum simulation and inverse parameter discovery within a continuous, mesh-free framework. While early benchmark investigations have demonstrated promising results in idealized open-channel flows with net through-flow, the applicability, accuracy, and optimization stability of PINNs in enclosed, recirculating geometries characterized by mixed kinematic regimes remain largely unexplored.

This dissertation investigates the formulation, performance, and physical identifiability boundaries of physics-informed neural network architectures applied to the four-roll mill geometry—a canonical hydrodynamic system featuring a central hyperbolic planar extensional stagnation point surrounded by intense shear layers adjacent to rotating cylinders.

1.1 Structure of the Dissertation

The remainder of this dissertation is organized into four sequential chapters:

- **Chapter 2 (Viscoelastic Fluid Mechanics and the Four-Roll Mill):**

Establishes the continuum mechanics foundation, governing Navier–Stokes equations, and constitutive closures for Oldroyd-B and Giesekus fluids. It formally characterizes the four-roll mill cavity, its kinematic flow classification, and the stream-function parameterization.

- **Chapter 3 (Physics-Informed Neural Networks in Complex Rheology):** Reviews deep feedforward neural architectures, automatic differentiation, and the foundational PINN paradigm. It contrasts traditional machine learning generalization with continuous PDE boundary-value solving, and surveys prior literature benchmarks in computational rheology.
- **Chapter 4 (Results and Discussion: Four-Roll Mill PINN Solver):** Formulates the stream-function PINN solver for the four-roll mill and details the staged optimization strategy. It presents rigorous forward validations against high-fidelity COMSOL FEM solutions, inverse parameter estimation across linear and non-linear rheologies, grid independence analyses, transfer learning stability, and the mathematical origin of PTT shear flow degeneracies.
- **Chapter 5 (Conclusions and Future Perspectives):** Synthesizes the core scientific findings of this research, reviews operational limitations regarding boundary stress supervision, and outlines promising future research directions.

Chapter 2

Viscoelastic Fluid Mechanics and the Four-Roll Mill

In this chapter, the continuum mechanics framework governing the flow of incompressible viscoelastic fluids is established. The primary objective is to formulate the governing equations and constitutive models that describe the physical system within the four-roll mill geometry. We begin from the universal conservation principles of mass and linear momentum, introducing the physical and kinematic simplifications characteristic of polymer solutions under creeping and low-inertia flow conditions. The Oldroyd-B constitutive model is formulated by examining the Upper-Convected Time Derivative (UCTD), detailing its extensional kinematics and the origins of the High-Weissenberg Number Problem (HWNP). Non-linear closures, specifically the Giesekus model (accounting for anisotropic hydrodynamic drag) and the Phan-Thien–Tanner (PTT) model (accounting for stress-dependent macromolecular entanglement breakdown), are presented to establish the theoretical continuum spectrum. Crucially, the active computational and PINN framework of this dissertation focuses specifically on the **Oldroyd-B** and **Giesekus** models, utilizing PTT as an exploratory theoretical counterpart in parameter identifiability analysis. A viscous stress scaling procedure is introduced

to define the governing dimensionless groups ($Re_{\text{scale}}, Re_{\text{phys}}, Wi, \beta$). Finally, the four-roll mill benchmark is formalized through its local kinematics, flow classification, geometry, diagnostic verification cutlines, boundary conditions, and the definition of the forward and inverse problems.

2.1 General Conservation Laws

From a continuum mechanics perspective, the spatial and temporal motion of a fluid continuum is governed by the universal conservation laws of mass and linear momentum [1].

2.1.1 Principle of Mass Conservation

The conservation of mass for a fluid with local density $\rho(\mathbf{x}, t)$ and velocity vector $\mathbf{u}(\mathbf{x}, t)$ is expressed through the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (2.1)$$

Equation (2.1) dictates that the instantaneous rate of mass accumulation within any fixed control volume must be strictly balanced by the net convective mass flux through its boundary surface.

2.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the momentum equation in Eulerian coordinates:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (2.2)$$

where $\mathbf{g}$ denotes the acceleration due to external body forces, and $\boldsymbol{\sigma}$ is the symmetric second-order stress tensor. The left-hand side represents the substantial derivative $D\mathbf{u}/Dt$, which combines the local unsteady rate of change $\partial\mathbf{u}/\partial t$ with the convective acceleration term $\mathbf{u} \cdot \nabla \mathbf{u}$ arising from spatial velocity gradients.

To isolate the isotropic pressure field from internal stresses generated by deformation, the total stress tensor is partitioned as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (2.3)$$

where p is the isotropic pressure field, $\mathbf{I}$ is the second-order identity tensor, and $\mathbf{T}$ is the extra-stress tensor. The extra-stress tensor encapsulates all non-hydrostatic stresses associated with the dissipative and elastic response of the fluid microstructure. Substituting Equation (2.3) into Equation (2.2) yields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (2.4)$$

2.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion can be reduced by introducing scale-appropriate physical assumptions for polymer solutions within the four-roll mill geometry:

1. **Continuous medium and state of rest:** The fluid is modeled as a continuous medium that is locally isotropic in its thermodynamic state of rest. Under kinematic deformation, flow-induced macromolecular orientation and stretching break this resting isotropy, generating stress anisotropies captured by the extra-stress tensor.

2. **Incompressibility and stream function formulation:** Under the moderate operating pressures and low flow velocities characteristic of liquid polymer systems, density variations are negligible ($\rho = \text{const}$). The mass conservation equation (2.1) simplifies to the divergence-free condition:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (2.5)$$

In two-dimensional planar motion, this incompressibility constraint can be satisfied identically by introducing a scalar stream function $\psi(x, y)$ defined such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (2.6)$$

By Schwarz's theorem on mixed partial derivatives, $\nabla \cdot \mathbf{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0$ holds identically across the entire domain.

3. **Isothermal operating conditions:** Under the operating conditions considered in this study, internal temperature variations due to viscous dissipation remain negligible. The energy conservation equation decouples from momentum, allowing transport properties (viscosities, relaxation time) to be treated as constants.
4. **Absence of body forces:** The four-roll mill geometry is analyzed in a horizontal cross-sectional plane (x, y) . Gravitational acceleration acts along the vertical cylinder axis (z) , perpendicular to the modeled planar flow field. Consequently, body force contributions vanish in the modeled plane, and we set $\mathbf{g} = \mathbf{0}$.
5. **Steady-state flow:** Once constant operating conditions are established and the flow reaches a steady regime, all local temporal derivatives vanish identically ($\partial(\cdot)/\partial t = \mathbf{0}$).

6. **Two-dimensional planar kinematics:** Given the high aspect ratio of the cylindrical rollers along the axial dimension (z -axis), edge effects in the transverse mid-plane of the cavity are negligible. The velocity field is therefore strictly planar:

$$\mathbf{u} = (u(x, y), v(x, y), 0). \quad (2.7)$$

Under these physical simplifications, the momentum equation reduces to:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot \mathbf{T}. \quad (2.8)$$

2.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (2.8) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, whereas mass and momentum conservation provide only three equations. A constitutive relationship for the extra-stress tensor $\mathbf{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, the extra-stress tensor $\mathbf{T}$ is partitioned into a purely viscous Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (2.9)$$

where μ_s represents the dynamic viscosity of the Newtonian solvent, and $\mathbf{D}$ denotes the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (2.10)$$

This decomposition isolates the instantaneous dissipative response of the solvent ($2\mu_s \mathbf{D}$) from the elastic stress $\boldsymbol{\tau}$ generated by macromolecular stretching and orientation. Substituting Equation (2.9) into Equation (2.8) and exploiting incompressibility ($\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$), the steady momentum equation becomes:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (2.11)$$

2.3.1 Upper-Convected Time Derivative

Polymeric liquids exhibit memory: macromolecular chains deform, orient, and relax over characteristic finite timescales. When a fluid element moves through a complex flow, its internal stress state is simultaneously advected along streamlines, rotated by local vorticity, and stretched by velocity gradients.

To satisfy the principle of material frame-indifference, constitutive equations must employ a frame-invariant tensor rate. For flexible polymer chains undergoing affine deformation with the surrounding continuum, the appropriate objective derivative is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$. With the velocity gradient defined as $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$, the upper-convected derivative of the symmetric extra-stress tensor $\boldsymbol{\tau}$ is:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T. \quad (2.12)$$

Under steady conditions, $\partial \boldsymbol{\tau} / \partial t = \mathbf{0}$. The term $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$ accounts for convective spatial transport of stress along flow streamlines, while the terms $-(\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T$ describe the stretching and rotation of macromolecular conformations induced by local velocity gradients.

2.3.2 The Oldroyd-B Model and Extensional Behavior

The classical Oldroyd-B constitutive model [4] represents macromolecules as dilute, non-interacting Hookean dumbbells suspended in a Newtonian solvent [1]. In dilute solution theory, the polymeric viscosity contribution μ_p is directly coupled to the relaxation time λ through kinetic theory:

$$\mu_p = nk_B T \lambda, \quad (2.13)$$

where n is the number density of polymer chains, k_B is the Boltzmann constant, and T is absolute temperature. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}. \quad (2.14)$$

The total zero-shear viscosity of the fluid is $\mu_0 = \mu_s + \mu_p$.

While Oldroyd-B predicts a constant shear viscosity $\eta(\dot{\gamma}) = \mu_0$ (characteristic of ideal elastic fluids), it exhibits an intrinsic limitation in extensional kinematics. In a steady planar extensional flow ($u = Ex, v = -Ey$, with extension rate $E > 0$), as the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \quad (2.15)$$

the assumption of infinitely extendable Hookean dumbbells allows unbounded stretching. Consequently, the polymeric stress and extensional viscosity diverge asymptotically to infinity ($\eta_E \rightarrow \infty$). At low deformation rates ($E \rightarrow 0$), the planar extensional viscosity recovers the Newtonian limit, yielding a planar Trouton ratio $\text{Tr}_E = \eta_E/\mu_0 \rightarrow 4$.

This unphysical stress divergence near the critical threshold $2\lambda E \rightarrow 1$ is the physical manifestation of the *High-Weissenberg Number Problem* (HWNP), which

causes severe numerical breakdown in conventional grid-based solvers due to exponentially steep stress boundary layers [3].

2.3.3 Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)

To overcome the unphysical stress singularity of Oldroyd-B and capture the behavior of semi-dilute solutions and concentrated polymer melts, non-linear constitutive models introduce stress-limiting mechanisms. The Giesekus and Phan-Thien–Tanner (PTT) models represent two distinct extensions of the baseline Oldroyd-B framework, incorporating different physical mechanisms to modify the rheological response at high deformation rates while retaining the same linear baseline at low strain rates:

The Giesekus Model

The Giesekus constitutive equation [1,20] accounts for anisotropic Brownian drag and hydrodynamic interactions between neighboring macromolecules in concentrated systems. This is achieved by introducing a quadratic stress relaxation term:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} \boldsymbol{\tau}^2 = 2\mu_p \mathbf{D}, \quad (2.16)$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and the tensorial square is defined by matrix contraction $(\boldsymbol{\tau}^2)_{ij} = \tau_{ik}\tau_{kj}$. When $\alpha = 0$, the model reduces to Oldroyd-B. For $\alpha > 0$, the quadratic term limits the growth of extensional stresses, preventing the unphysical divergence of Oldroyd-B and introducing shear-thinning behavior in steady shear flows.

The Phan-Thien–Tanner (PTT) Model

Derived from the transient network theory of concentrated polymeric systems [1,21], the PTT model assumes that polymer chains form temporary physical entanglements whose rate of creation and destruction depends on the stored elastic stress. In the linear PTT formulation, the constitutive equation is:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}, \quad (2.17)$$

where the stress-softening function $f(\boldsymbol{\tau})$ depends linearly on the trace of the extra-stress tensor:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}), \quad (2.18)$$

with $\text{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$, and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter controlling the stress-induced rate of entanglement breakdown. When $\varepsilon = 0$, $f = 1$ and the model recovers Oldroyd-B. For $\varepsilon > 0$, the effective relaxation rate increases with stress, bounding the extensional viscosity.

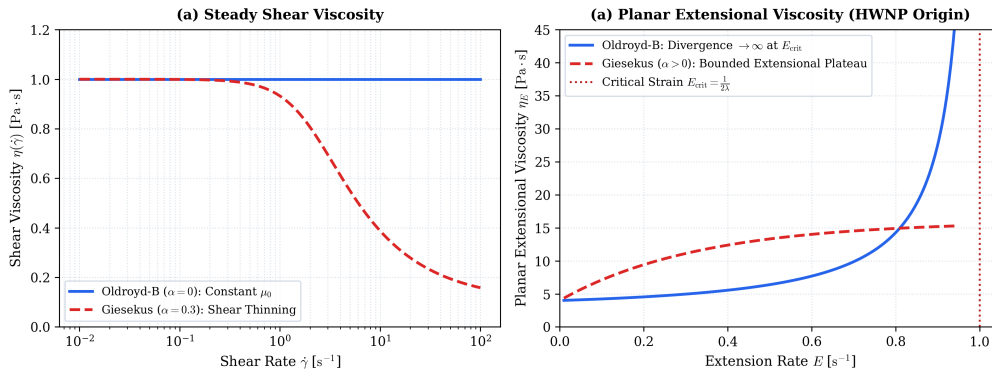


Figure 2.1: Conceptual comparison of constitutive model responses: (a) steady shear viscosity $\eta(\dot{\gamma})$, highlighting the constant viscosity of Oldroyd-B versus the shear-thinning behavior of Giesekus ($\alpha > 0$); (b) planar extensional viscosity $\eta_E(E)$, demonstrating the asymptotic stress divergence of Oldroyd-B at the critical strain rate $E_{\text{crit}} = 1/(2\lambda)$ (origin of the High-Weissenberg Number Problem) versus the bounded extensional plateau captured by the non-linear Giesekus formulation.

Focus of Present Investigation: Oldroyd-B and Giesekus Formulations

While the broader rheological literature encompasses a multitude of constitutive equations, this dissertation intentionally focuses its computational and scientific investigation on two representative viscoelastic archetypes:

1. **The Oldroyd-B Model:** Serves as the fundamental linear benchmark of constant-viscosity elastic liquids (Boger fluids), embodying the purest manifestation of elastic normal stress growth and the High-Weissenberg Number Problem.
2. **The Giesekus Model:** Provides the primary non-linear extension, incorporating molecular mobility ($\alpha \in (0, 0.5]$) to capture shear-thinning and bounded extensional stresses characteristic of semi-dilute solutions.

The linear PTT formulation is maintained in the governing differential expressions for theoretical generality, but is set aside ($\varepsilon = 0$) during primary benchmarks, serving later as an exploratory case for examining parameter identifiability limits in recirculating shear regimes.

2.4 Nondimensionalization and Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

2.4.1 Reference Scales and Dimensionless Groups

We select reference length H_{ref} (roller radius R), reference velocity U_{ref} (roller surface speed u_{roll}), and a constant reference viscosity η_0 (chosen as a fixed baseline scale, e.g., $\eta_0 = 1.0 \text{ Pa} \cdot \text{s}$). The dimensionless variables are defined as:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (2.19)$$

Dropping asterisks for notational brevity, the governing non-dimensional groups are:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (2.20)$$

$$Re_{\text{phys}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\mu_s + \mu_p} = \frac{Re_{\text{scale}}}{\mu_s^* + \mu_p^*} \quad (\text{Physical Reynolds number}), \quad (2.21)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Global Weissenberg number}), \quad (2.22)$$

$$\beta = \frac{\mu_s}{\mu_s + \mu_p} = \frac{\mu_s^*}{\mu_s^* + \mu_p^*} \quad (\text{Viscosity ratio}), \quad (2.23)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless solvent and polymer viscosities}). \quad (2.24)$$

The distinction between Re_{scale} and Re_{phys} is essential in inverse problems: defining the momentum equation through the constant scale Re_{scale} ensures that the numerical scaling of convective acceleration remains invariant even as unknown fluid viscosities (μ_s, μ_p) are adjusted during optimization.

2.4.2 Dimensionless Vector Formulation

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (2.25)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (2.26)$$

$$\mathcal{G}(\boldsymbol{\tau}) + Wi (\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T) - 2\mu_p^* \mathbf{D} = \mathbf{0}, \quad (2.27)$$

where the constitutive operator $\mathcal{G}(\boldsymbol{\tau})$ unifies the models under study:

$$\mathcal{G}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B),} \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*} \boldsymbol{\tau}^2 & \text{(Giesekus),} \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*} \text{tr}(\boldsymbol{\tau})\right) \boldsymbol{\tau} & \text{(Linear PTT).} \end{cases} \quad (2.28)$$

2.5 The Four-Roll Mill Benchmark Kinematics and Boundaries

The four-roll mill is a canonical benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally conceived by Taylor [5] and systematically developed by Fuller and Leal [7, 17], it has served as a standard apparatus for analyzing macromolecular deformation and flow-induced birefringence. As depicted in Figure 2.2, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

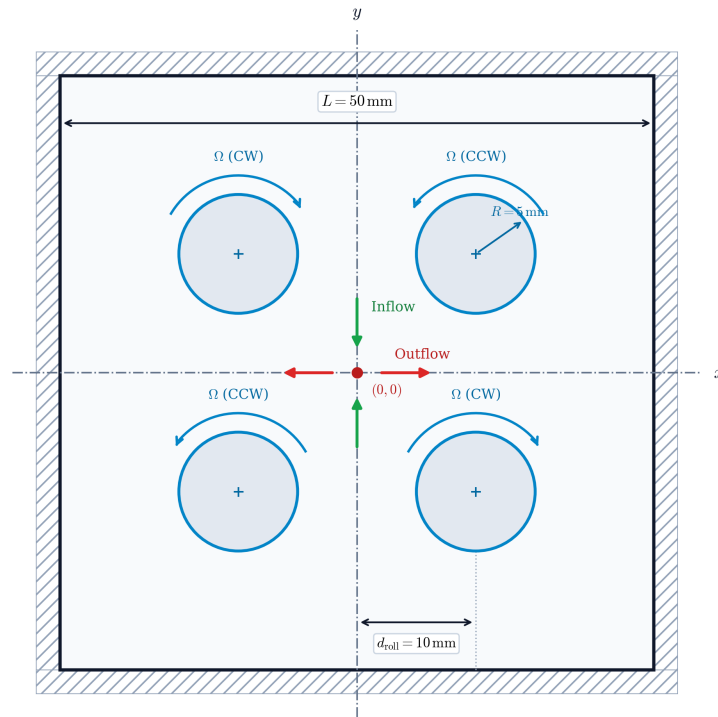


Figure 2.2: Schematic representation of the four-roll mill geometry, illustrating the square cavity of side L , cylindrical rollers of radius R at offset d_{roll} , Cartesian axes (x, y) , rotation directions with angular velocity Ω (adjacent rolls counter-rotate, diagonal rolls co-rotate), and the central hyperbolic stagnation point $(0, 0)$.

2.5.1 Kinematic Relations and Flow Classification

In the vicinity of the central region between the rollers, the local velocity field can be represented by a linear superposition of extension and rotation [7, 17]:

$$\mathbf{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \quad (2.29)$$

where $E \geq 0$ denotes the extension rate and $\omega \geq 0$ represents the local angular rotation rate. The associated scalar vorticity of this field is $\omega_z = (\nabla \times \mathbf{u})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2\omega$. To quantify the relative magnitude of extension versus rotation, a dimensionless flow type parameter $\chi \in [-1, 1]$ is defined:

$$\chi = \frac{E - \omega}{E + \omega}. \quad (2.30)$$

Depending on the roller speeds and rotation directions, this idealized local kinematics captures three canonical flow regimes (illustrated in Figure 2.3):

- $\chi = 1$ ($\omega = 0$): **Pure Planar Extensional Flow.** Generated when adjacent rolls counter-rotate symmetrically (diagonal rolls co-rotate). Fluid elements experience pure stretching along x and pure compression along y , with hyperbolic streamlines $xy = \text{const}$.
- $\chi = 0$ ($E = \omega$): **Simple Shear Flow.** An intermediate state where rate of strain and angular rotation have equal magnitude ($E = \omega$), producing rectilinear shear streamlines. In an enclosed cavity, this uniform shear field is realized locally near the center, while further away the flow curves to adapt to roller and wall boundaries.
- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation.** Generated when rolls co-rotate to induce bulk circulation, yielding concentric circular streamlines

around the origin representing rigid-body rotation without elongation or compression.

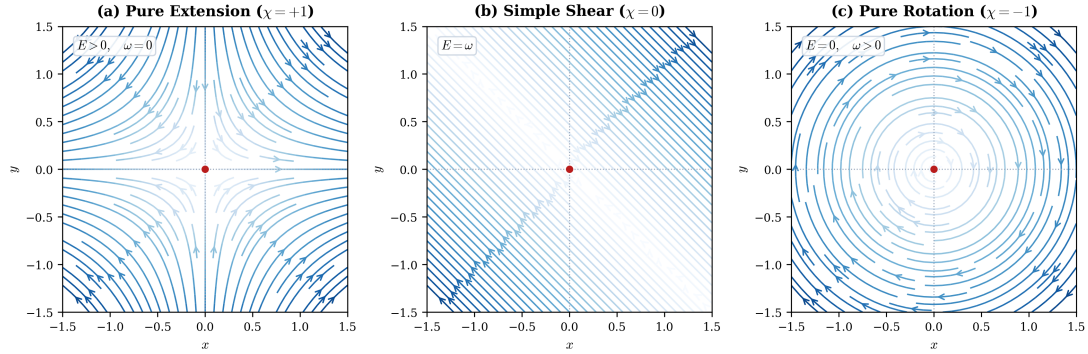


Figure 2.3: Streamline patterns of the two-dimensional linear flow field in Equation (2.29) for the three canonical regimes of the flow classification parameter χ : (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

2.5.2 The Central Hyperbolic Stagnation Point

In the extensional regime ($\chi = 1$), the velocity vanishes at the geometric center:

$$\mathbf{u}(0, 0) = \mathbf{0}. \quad (2.31)$$

The velocity gradient tensor at the origin,

$$\nabla \mathbf{u} = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}, \quad (2.32)$$

possesses real eigenvalues $+E$ (along the horizontal stretching axis x) and $-E$ (along the vertical compression axis y), defining a hyperbolic saddle-point geometry. Along the incoming vertical streamline ($x = 0$), fluid parcels decelerate toward zero velocity, providing fluid elements with long residence times.

This kinematic feature allows macromolecular deformation to accumulate over timescales comparable to or larger than the fluid relaxation time λ , providing an ideal testbed for observing elastic stress growth away from solid boundaries.

2.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D’Avino et al. [18], the physical configuration is characterized by:

- **Cavity dimensions:** Square enclosure of side $L = 50$ mm spanning $[-25, 25]$ mm;
- **Cylindrical rollers:** Radius $R = 5$ mm ($H_{\text{ref}} = R = 0.1L$);
- **Roller placement:** Centers positioned symmetrically at $(\pm 10, \pm 10)$ mm ($d_{\text{roll}} = 10$ mm);
- **Roller kinematics:** Peripheral surface velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}$ m/s;
- **Non-dimensional domain:** In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

The physical and operating parameters investigated in this work are summarized in Table 2.1.

2.5.4 Diagnostic Verification Cutlines

To enable rigorous, high-resolution validation of kinematic and stress profiles across the domain, three representative diagnostic cutlines are established (Figure 2.4):

Table 2.1: Physical and operating parameters for the four-roll mill benchmark.

Parameter	Symbol	Value / Range	Unit
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller surface velocity	u_{roll}	8.333×10^{-3}	m/s
Fluid density	ρ	1000	kg/m ³
Characteristic viscosity scale	η_0	1.0	Pa · s
Polymer relaxation time	λ	0.05–1.0	s
Viscosity ratio	β	0.1–0.5	—
Scaling Reynolds number	Re_{scale}	$0.0417 \ll 1$	—
Weissenberg number	Wi	0.083–1.667	—

1. **Horizontal Cutline** ($y = 0, -L/2 \leq x \leq L/2$): Traversing the central extensional outflow axis through the hyperbolic stagnation point $(0, 0)$. Along this path, fluid accelerates outward, forming intense elongational stress tails.
2. **Vertical Cutline** ($x = 0, -L/2 \leq y \leq L/2$): Traversing the compressive inflow axis, capturing fluid deceleration into the central stagnation zone.
3. **Oblique Diagonal Cutline** ($y = x$): Connecting opposite cavity corners and passing directly through the narrow roller nip gaps, where steep localized shear and velocity gradients reach their physical peaks.

These diagnostic trajectories serve as the spatial evaluation baselines for both the reference CFD mesh convergence study and the PINN field validation detailed in Chapter 4.

2.5.5 Boundary Conditions and Pressure Point Anchoring

To complete the continuous boundary value formulation, physical boundary conditions must be specified along all solid surfaces:

1. **Tangential Velocity on Roller Walls** (Γ_{rolls}): On the surface of each roller $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies no-slip with the

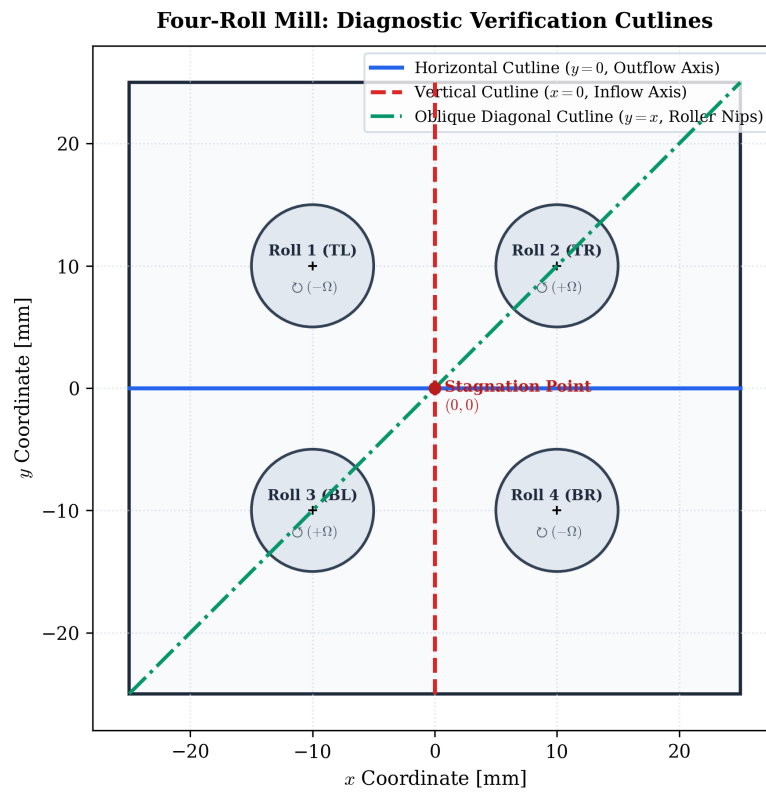


Figure 2.4: Schematic of the four-roll mill geometry depicting the four counter-rotating cylinders, the central extensional stagnation point (0,0), and the three diagnostic verification cutlines: Horizontal ($y = 0$, blue), Vertical ($x = 0$, red dashed), and Oblique Diagonal ($y = x$, green dash-dot).

prescribed peripheral rotation:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (2.33)$$

where $\mathbf{x}_k$ is the roller center. For the planar extensional regime ($\chi = 1$), adjacent rolls counter-rotate while diagonal rolls co-rotate: top-left and bottom-right rolls rotate clockwise ($-\Omega$), while top-right and bottom-left rolls rotate counter-clockwise ($+\Omega$).

2. **No-Slip Velocity on Outer Cavity Walls** (Γ_{outer}): Along the stationary external boundaries of the square enclosure, the velocity vanishes:

$$\mathbf{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \quad (2.34)$$

3. **Pressure Point Anchoring**: Because pressure appears in the momentum balance (2.26) exclusively through its gradient ∇p , in an enclosed cavity with prescribed velocity Dirichlet boundary conditions the pressure is determined only up to an arbitrary additive constant ($p \rightarrow p + C$). To fix this gauge freedom and obtain a unique solution, a single reference pressure point is specified:

$$p(\mathbf{x}_0) = 0, \quad (2.35)$$

at an arbitrary reference point $\mathbf{x}_0$ within the domain.

2.6 Problem Formulation and Computational Transition

The physical laws, constitutive closures, and boundary constraints established in this chapter define two complementary problem settings:

2.6.1 Forward and Inverse Problem Formulations

- **Forward (Direct) Problem:** Given a complete set of known fluid parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ and the prescribed kinematic boundary conditions on rolls and walls, determine the velocity field $\mathbf{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy the momentum balance (2.26) and constitutive closure (2.27) throughout the cavity domain Ω .
- **Inverse (Parameter Identification) Problem:** Given partial kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields $(p, \boldsymbol{\tau})$ and identify the unknown physical parameters governing the fluid $(\lambda, \mu_p, \mu_s, \text{alongside non-linear parameters } \alpha \text{ or } \varepsilon)$.

Table 2.2 classifies the role of each variable and parameter across the two problem formulations.

Table 2.2: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	Symbol	Forward Problem	Inverse Problem
Velocity Field	$\mathbf{u} = (u, v)$	Primary unknown (via ψ)	Observed / Supervised
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	$\lambda (Wi)$	Known parameter	Unknown physical parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown physical parameter
Solvent Viscosity	μ_s	Known parameter	Unknown physical parameter
Giesekus Mobility	α	Known parameter	Unknown parameter / Model selection
PTT Extensibility	ε	Known parameter	Unknown parameter / Model selection

2.6.2 Transition to Physics-Informed Neural Networks

The differential system formulated in Equations (2.25)–(2.28) represents a highly coupled, non-linear system of partial differential equations. In Chapter 3, this continuous physical formulation is translated into a machine-learning framework using Physics-Informed Neural Networks (PINNs). By embedding the momentum and constitutive equations directly into the neural network loss function

through automatic differentiation, the PINN framework enables both forward flow reconstruction and inverse rheological parameter discovery without requiring traditional spatial grids.

Chapter 3

Physics-Informed Neural Networks in Complex Rheology

Feedforward deep networks, commonly known as Multi-Layer Perceptrons (MLPs), represent the quintessential architecture of deep learning [15]. Conceptually, an MLP is a parametric function defined by composing together a sequence of simpler parametric functions. Each of these constituent functions has multiple inputs and outputs, and in neural network terminology is called a *layer*. The scalar outputs computed within each layer are referred to as *units* or *features*. Although each unit performs a relatively simple mathematical mapping of its inputs, chaining them in succession allows the network as a whole to express continuous functions of arbitrary complexity.

The structural dimensions of a network are conventionally characterized by its *width*—the number of units in a given layer—and its *depth*—the total number of layers. Having significant width and depth provides deep architectures with remarkable representational power, making them the foundational technology behind modern machine learning applications.

To appreciate the paradigm shift introduced by deep neural networks, it is useful to contrast them with traditional machine learning algorithms, such as

linear regression, logistic classifiers, and kernel methods. Classical algorithms operate by applying a linear transformation to a fixed set of features. These methods can indeed learn non-linear behaviors, but their non-linearity is rigid: it is non-linear only in the original input space, while remaining strictly linear in a feature space that must be pre-defined and engineered manually by human experts. Neural networks fundamentally overcome this limitation by automating the design of features. Rather than relying on manual feature engineering, an MLP learns the optimal representations directly from data.

In this architecture, the layers between the inputs and the final outputs are termed *hidden layers*. They are called hidden because their internal values are not specified or observed in the training data; instead, the optimization algorithm must autonomously determine how to organize these intermediate representations to solve the target problem. By transforming inputs through successive hidden layers, neural networks can approximate complex continuous relationships.

In this chapter, we explore how these feedforward architectures form the computational core of scientific machine learning. We begin by reviewing the fundamentals of neural networks and their optimization. We then examine why continuous neural representations are compelling for physical modeling, while discussing the critical limitations of purely data-driven black-box approaches. This analysis naturally motivates the Physics-Informed Neural Network (PINN) paradigm, followed by a review of foundational literature benchmarks and their applications in complex non-Newtonian flows.

3.1 Fundamentals of Deep Learning

Deep learning algorithms approximate continuous mathematical functions through multi-layer architectures that learn internal representations from data [15]. In this section, we present the essential building blocks of these networks, focus-

ing on their mathematical structure, smooth activation functions, optimization algorithms, and gradient evaluation.

3.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary processing unit of a neural network is the artificial neuron. Given an input vector $\mathbf{x}$, a neuron computes an affine combination of the inputs followed by a scalar non-linear mapping:

$$z = \mathbf{w} \cdot \mathbf{x} + b, \quad a = \sigma(z), \quad (3.1)$$

where $\mathbf{w}$ is the weight vector, b is a scalar bias, and $\sigma(\cdot)$ is the activation function. The activation function introduces non-linearity into the model. Without non-linear activations, composing multiple linear layers would simply collapse mathematically into another single linear mapping, eliminating the expressive advantage of deep architectures.

In standard deep learning applications such as image recognition, piecewise-linear activations—most notably the Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$)—are widely adopted due to their computational simplicity. However, in scientific computing and fluid mechanics, the network’s outputs represent continuous physical fields governed by partial differential equations (PDEs), such as velocities, pressures, and stresses. These governing laws involve spatial derivatives of first, second, or higher order (such as viscous laplacians $\nabla^2 \mathbf{u}$ or stress divergences $\nabla \cdot \boldsymbol{\tau}$).

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere ($\text{ReLU}''(z) \equiv 0$), evaluating a ReLU network inside differential equations would yield zero for diffusive and curvature terms. For this reason, physics-informed models strictly require smooth, infinitely differentiable

activation functions ($\mathcal{C}^\infty$). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [15]:

$$\text{SiLU}(z) = \frac{z}{1 + e^{-z}}. \quad (3.2)$$

The SiLU activation is smooth, non-monotonic, and self-gated. Unlike classical functions such as the hyperbolic tangent ($\tanh$) that saturate toward constant limits for large inputs, SiLU remains active in the positive regime. This property helps maintain healthy gradient flow across regions characterized by intense localized shear and steep stress variations, which are typical of viscoelastic flows.

3.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector $\mathbf{x}$ to an output prediction $\hat{\mathbf{u}}$ by propagating information forward through L successive layers:

$$\begin{aligned} \mathbf{h}^{(1)} &= \sigma \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right), \\ \mathbf{h}^{(l)} &= \sigma \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad \text{for } l = 2, \dots, L-1, \\ \hat{\mathbf{u}} &= \mathbf{W}^{(L)} \mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \end{aligned} \quad (3.3)$$

where $\mathbf{h}^{(l)}$ denotes the hidden feature vector at layer l , and $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ represent the weight matrix and bias vector of that layer. In the context of this thesis, the inputs $\mathbf{x} = (x, y)$ are spatial coordinates within the fluid domain, while the outputs $\hat{\mathbf{u}}$ correspond to the continuous physical fields of interest, such as the stream function ψ , the pressure p , and the extra-stress tensor components $\boldsymbol{\tau}$.

A central theoretical property of MLPs is their universal approximation capability [14, 23]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a closed domain to an

arbitrary degree of accuracy.

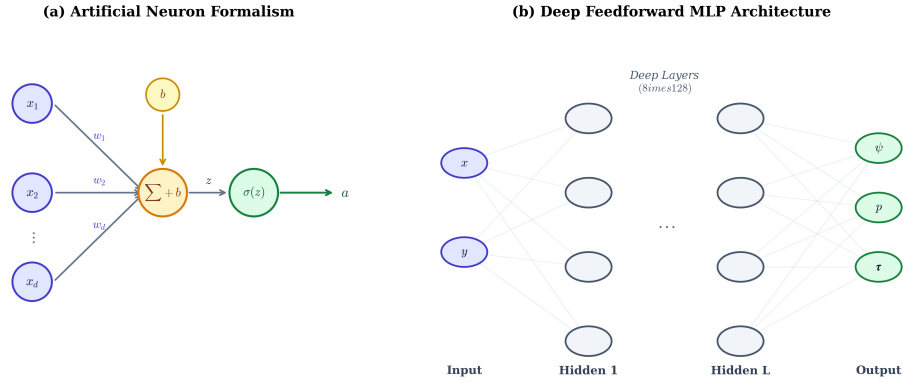


Figure 3.1: Foundational deep learning representations: (a) schematic of an individual artificial neuron computing an affine combination $z = \mathbf{w} \cdot \mathbf{x} + b$ followed by a smooth non-linear activation $\sigma(z)$; (b) deep Multi-Layer Perceptron (MLP) architecture propagating coordinate inputs (x, y) through L hidden layers to parameterize continuous physical fields $(\psi, p, \boldsymbol{\tau})$.

Weight Initialization: Glorot Scheme and Output Zero-Initialization

The initialization of the network weights and biases strongly influences optimization stability. To avoid vanishing or exploding signals across deep layers, the hidden layer weights are initialized using the Xavier/Glorot normal distribution [15], which scales the variance of the initial parameters according to the width of adjacent layers.

For the final output layers representing the extra-stress tensor $\boldsymbol{\tau}$ and the pressure p , we adopt a strategic zero-initialization ($\mathbf{W}^{(L)} = \mathbf{0}, \mathbf{b}^{(L)} = \mathbf{0}$). This choice is motivated by two practical reasons:

1. **Thermodynamic Baseline:** In fluid mechanics, a fluid at thermodynamic rest has zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$). Zero-initialization guarantees that at iteration zero, the network outputs correspond to the unperturbed resting state.
2. **Suppression of Numerical Shocks:** If output weights were randomly

initialized, initial stress predictions would exhibit high-frequency spatial noise across the domain. Evaluating these random fluctuations inside non-linear constitutive equations—which contain quadratic stress products and convective gradients—would cause initial residual losses to spike violently, destabilizing the optimization. Initializing the output layer to zero ensures smooth, bounded initial residuals.

3.1.3 Loss Functions and Optimization Algorithms

Training a neural network consists of finding the optimal set of parameters $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}$ that minimizes a scalar loss function $\mathcal{L}(\boldsymbol{\theta})$. The loss measures how well the network’s predictions match desired targets, typically using the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{u}}(\mathbf{x}_i; \boldsymbol{\theta}) - \mathbf{u}_i^{\text{target}}\|_2^2. \quad (3.4)$$

The parameters are updated iteratively in the direction that reduces the loss, following the principle of gradient descent:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}, \quad (3.5)$$

where $\eta > 0$ is the learning rate, which controls the step size taken along the negative gradient.

In this work, training is carried out using a two-stage optimization strategy:

1. **First-Order Optimization (Adam):** The Adaptive Moment Estimation optimizer [22] dynamically adjusts the learning rate for each individual parameter based on running estimates of past gradients. This makes Adam robust and effective at exploring complex loss landscapes during the early stages of training.

2. **Second-Order Quasi-Newton Optimization (L-BFGS):** Once the solution is close to a minimum, the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm uses curvature information to achieve faster, super-linear convergence. In scientific problems, L-BFGS is crucial for refining the solution and driving physical residuals down to high accuracy ($10^{-5} - 10^{-7}$).

3.1.4 Backpropagation and Computational Graphs

To update the parameters via gradient descent, the network must compute the derivative of the loss with respect to every weight and bias, denoted by $\nabla_{\theta}\mathcal{L}$. Evaluating these derivatives numerically by perturbing each parameter one by one would be computationally infeasible.

Instead, neural networks compute exact gradients efficiently using the *backpropagation* algorithm [15]. The execution of a neural network can be viewed as a directed computational graph, where inputs flow forward through mathematical operations to produce the predictions and the final loss value. Backpropagation applies the classical chain rule of calculus in reverse order, starting from the loss and propagating sensitivities backward through the graph.

Through this reverse-mode evaluation, the gradients with respect to all parameters in the network are computed simultaneously in a single backward pass, at a computational cost roughly equivalent to that of a single forward evaluation. This computational efficiency is what makes training deep models practical.

3.2 Why Neural Networks? Capabilities and Physical Limitations

3.2.1 Capabilities and Universal Function Approximation

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **Continuous and Meshless Representation:** Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a continuous, globally defined mapping $\mathbf{x} \mapsto \hat{\mathbf{u}}(\mathbf{x})$ over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being meshless does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.
2. **High-Dimensional Functional Approximation:** The universal approximation capacity allows neural networks to represent complex, coupled physical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.
3. **Hardware Acceleration and Differentiability:** Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

3.2.2 Critical Bottlenecks of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven, black-box neural networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

- **Severe Data Scarcity in Experimental Rheology:** Deep neural networks conventionally rely on extensive labeled datasets. In experimental fluid mechanics, acquiring dense velocity fields via techniques such as Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are generally unavailable in complex geometries.
- **Violation of Fundamental Conservation Laws:** Purely data-driven models minimize statistical discrepancy over discrete training points without enforcing physical principles. Consequently, their predictions routinely violate mass conservation ($\nabla \cdot \mathbf{u} \neq 0$), linear momentum balance, and constitutive relationships, leading to unphysical artifacts—particularly when interpolating across sparse regions or extrapolating beyond observed conditions.
- **Overfitting and Spurious Oscillations:** In low-data regimes, over-parameterized networks can readily fit experimental noise or sparse sensor readings with minimal training error, while developing high-frequency, non-physical oscillations in unmonitored subdomains.
- **Inability to Identify Governing Parameters:** Black-box architectures fit mapping functions between inputs and outputs, but cannot infer underlying physical material properties—such as fluid relaxation times or polymeric viscosities—that govern the flow dynamics.

These fundamental shortcomings emphasize that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed the governing physical equations directly into the learning process.

3.3 The Physics-Informed Neural Network Paradigm

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [8], addresses the limitations of purely data-driven modeling by embedding mathematical governing equations directly into the objective function of a neural network.

3.3.1 Physics-Constrained Loss Formulation

Consider a general non-linear differential operator $\mathcal{F}$ governing a continuous physical state $\mathbf{u}(\mathbf{x})$ over a spatial domain $\Omega \subset \mathbb{R}^d$ bounded by $\partial\Omega$:

$$\mathcal{F}[\mathbf{u}(\mathbf{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad (3.6)$$

$$\mathcal{B}[\mathbf{u}(\mathbf{x})] = \mathbf{g}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (3.7)$$

where $\boldsymbol{\lambda}$ represents physical parameters (such as fluid viscosities or relaxation times) and $\mathcal{B}$ denotes boundary differential operators.

In a PINN, the unknown solution $\mathbf{u}(\mathbf{x})$ is approximated by a neural network $\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta})$. Substituting $\hat{\mathbf{u}}$ into the governing equations defines the differential residual:

$$\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \quad (3.8)$$

The network parameters $\boldsymbol{\theta}$ —and, in inverse problems, the unknown physical parameters $\boldsymbol{\lambda}$ —are optimized by minimizing a composite multi-objective loss func-

tion:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (3.9)$$

where:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \|\hat{\mathbf{u}}(\mathbf{x}_i^{\text{data}}) - \mathbf{u}_i^{\text{data}}\|_2^2, \quad (3.10)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \|\mathcal{B}[\hat{\mathbf{u}}(\mathbf{x}_j^{\text{bc}})] - \mathbf{g}(\mathbf{x}_j^{\text{bc}})\|_2^2, \quad (3.11)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \|\mathbf{f}(\mathbf{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda})\|_2^2. \quad (3.12)$$

Here, $\{\mathbf{x}_i^{\text{data}}\}$ denotes available experimental or numerical measurements, $\{\mathbf{x}_j^{\text{bc}}\}$ represents points sampled along physical boundaries $\partial\Omega$, and $\{\mathbf{x}_k^{\text{pde}}\}$ designates interior *collocation points* scattered across Ω .

This formulation naturally accommodates two distinct problem settings:

- **Forward Problems:** The physical parameters $\boldsymbol{\lambda}$ are fully known. The network optimizes its weights $\boldsymbol{\theta}$ to solve for the unknown fields $\hat{\mathbf{u}}(\mathbf{x})$, acting as a continuous PDE solver.
- **Inverse Problems:** One or more physical parameters in $\boldsymbol{\lambda}$ are unknown. The network simultaneously reconstructs the physical fields and identifies the unknown parameters by leveraging both available observations $\mathcal{L}_{\text{data}}$ and the physical consistency constraints $\mathcal{L}_{\text{pde}}$.

A critical distinction must be drawn between reconstructing physical fields and identifying governing parameters. A neural network may achieve low training residuals and appear to fit observed velocity fields accurately, yet the inferred parameters may remain inaccurate or poorly identified due to parameter correlations, insufficient kinematic variation in the data, or ill-conditioned loss landscapes. Integrating governing equations provides necessary structure, but does

not automatically guarantee parameter identifiability or solution uniqueness.

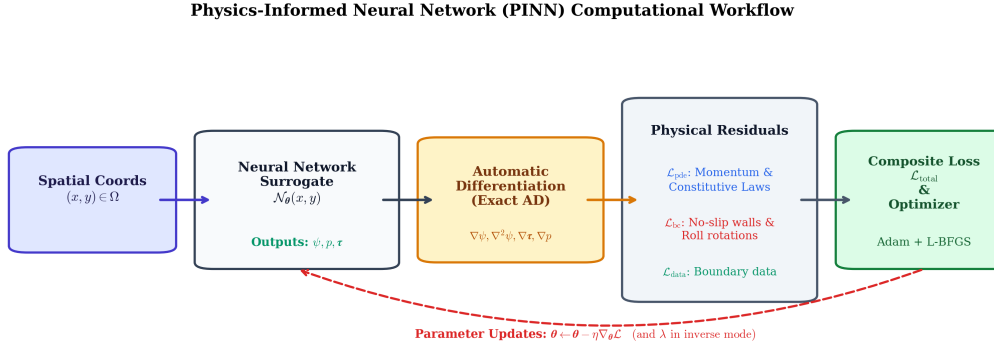


Figure 3.2: The Physics-Informed Neural Network (PINN) computational architecture: spatial coordinates (x, y) are propagated through a deep neural network surrogate, exact coordinate derivatives are evaluated via Automatic Differentiation (AD) without grid stencils, and parameters θ (and unknown physics λ) are updated iteratively by minimizing a multi-objective composite loss comprising PDE residuals, boundary conditions, and observed data.

3.3.2 Deep Dive: Standard Supervised Training vs. PINN Optimization

The operational differences between conventional supervised learning and physics-informed optimization are characterized by several key aspects:

1. **Training as a Continuous Boundary-Value Solver vs. Statistical Generalization:** In classical machine learning, a neural network is optimized on a finite training partition with the primary objective of statistical out-of-sample generalization to unseen test distributions $(X_{\text{test}} \rightarrow Y_{\text{test}})$. In contrast, in Physics-Informed Neural Networks, *the training process itself is the numerical solver*. The network is not designed as an inductive predictor to infer flow fields across disparate, unseen geometries; rather, the optimization of parameters θ directly computes the solution of a specific boundary-value problem on the domain Ω . Fitting empirical data $\mathcal{L}_{\text{data}}$ dur-

ing training does not function as an external test of generalization, but acts as an active physical regularization and data-assimilation mechanism that guides the optimization manifold toward the true physical state alongside differential residuals $\mathcal{L}_{\text{pde}}$ and boundary conditions $\mathcal{L}_{\text{bc}}$.

2. Physical Regularization and Inductive Bias: In standard supervised training, the network is unconstrained in regions devoid of training data, which allows over-parameterized models to fit noisy samples while producing spurious oscillations. In PINNs, evaluating the differential residual $\mathcal{L}_{\text{pde}}$ across interior collocation points imposes an analytical inductive bias: functions exhibiting non-physical gradients or curvature incur substantial penalties from the governing differential operators. This physical regularization significantly constrains the admissible solution space. However, it should not be viewed as an absolute guarantee against overfitting: if measurement data are noisy or if loss weights $w_{\text{data}}, w_{\text{pde}}$ are poorly balanced, the optimization can still overfit local measurements or converge to unphysical local minima.

3. Coordinate Automatic Differentiation (AD): In conventional deep learning, backpropagation employs AD to differentiate the scalar loss with respect to network parameters:

$$\text{Backpropagation (Classical DL): } \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}. \quad (3.13)$$

In PINNs, the exact same reverse-mode automatic differentiation engine is applied to compute spatial derivatives of the predicted state fields with respect to input coordinates:

$$\text{Coordinate AD (PINNs): } \frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{x}}, \quad \frac{\partial^2 \hat{\mathbf{u}}}{\partial \mathbf{x}^2}. \quad (3.14)$$

As summarized in Table 3.1, coordinate AD evaluates spatial derivatives analytically along the computational graph, avoiding the truncation error $\mathcal{O}(\Delta x^2)$ inherent in finite-difference stencils. Nonetheless, coordinate AD remains subject to machine floating-point roundoff, network functional approximation errors, and potential numerical ill-conditioning in regions of steep gradient transitions.

4. **Meshless Nature and Point Sampling:** Unlike traditional CFD techniques that rely on conforming discretization meshes and element connectivity tables, PINNs evaluate differential residuals at unorganized collocation point sets $\{\mathbf{x}_k^{\text{pde}}\}$. This meshless nature simplifies point generation in non-trivial domain boundaries. However, it replaces mesh generation with the challenge of point sampling: inadequate collocation density in boundary layers or near stagnation zones can prevent the network from resolving localized physical features.

Table 3.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	$\frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}$	$\mathcal{O}(\Delta x^2)$	Strict Structured Grid
Symbolic Differentiation	Closed-form analytical expression	None	Expression Swell
Automatic Differentiation (AD)	Exact chain rule on DAG	None (Machine Precision)	Mesh-Free / Continuous

3.4 Foundational Literature Benchmarks

3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational work, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capability of PINNs on the canonical benchmark of two-dimensional unsteady laminar flow past a circular cylinder at Reynolds number $Re = 100$, featuring periodic vortex shedding (the Von Kármán vortex street).

Problem Formulation and Inverse Discovery

The non-dimensional incompressible Navier–Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vv_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \quad (3.15)$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \quad (3.16)$$

$$u_x + v_y = 0, \quad (3.17)$$

where λ_1 represents the convective acceleration coefficient ($\lambda_1^{\text{true}} = 1.0$) and λ_2 corresponds to the kinematic viscosity ($\lambda_2^{\text{true}} = 1/Re = 0.01$).

To satisfy incompressibility (3.17) identically, Raissi et al. parameterized velocities via the scalar stream function formulation $\psi(t, x, y)$ through the exact relations $u = \partial\psi/\partial y, v = -\partial\psi/\partial x$ established in Eq. (2.6). A single neural network with 9 hidden layers and 20 neurons per layer, activated by hyperbolic tangent (tanh) functions, was trained to predict the latent fields (ψ, p) from coordinates (t, x, y) while treating λ_1 and λ_2 as unknown learnable parameters.

Latent Field Discovery and Parameter Identification

The network was trained on $N = 5,000$ spatio-temporal velocity observations sampled within the vortex street wake. This study highlighted two significant capabilities:

- **Latent Pressure Field Discovery:** Without any pressure measurements during training, the network reconstructed the spatio-temporal pressure field $p(t, x, y)$ solely through momentum residual enforcement, recovering the reference CFD pressure field up to an arbitrary hydrostatic constant.
- **Parameter Estimation Under Noise:** On clean velocity data, the authors reported relative errors of 0.08% for λ_1 and 4.67% for λ_2 . When 1%

uncorrelated Gaussian noise was introduced, the errors remained controlled at 0.17% for λ_1 and 5.70% for λ_2 .

While these results demonstrated the feasibility of simultaneous field reconstruction and parameter identification, the problem was governed by linear viscous stresses in a Newtonian fluid with open boundaries, leaving open the question of how such approaches behave under non-Newtonian constitutive equations.

3.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending the physics-informed framework to non-Newtonian media, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, demonstrating the application of PINNs to polymeric flows governed by differential constitutive models.

The Rheological Bottleneck

Viscoelastic fluid modeling presents severe numerical and experimental challenges:

1. **Experimental Limits:** Non-intrusive measurements of the internal extra-stress tensor $\boldsymbol{\tau}$ are generally unavailable in complex geometries, preventing full empirical characterization of stress fields.
2. **Numerical Breakdown (HWNP):** In conventional grid-based solvers (FEM, FVM), elevated elasticity triggers the High Weissenberg Number Problem (HWNP), where hyperbolic stress advection induces steep gradients that lead to numerical instability and loss of convergence, often requiring artificial stabilization techniques.

The Virtual Rheometer Concept

To address these challenges, ViscoelasticNet proposed an inverse framework conceptualized as a "virtual rheometer":

- **Latent Stress Inference:** Given velocity observations $\mathbf{u}$, the framework infers the unmeasured extra-stress tensor $\boldsymbol{\tau}$ and pressure field p by enforcing constitutive and momentum residuals.
- **Unified Constitutive Formulation:** The authors formulated a unified constitutive law encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}) \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (3.18)$$

seeking to identify the relaxation time λ , polymeric viscosity μ_p , and non-linear mobility parameters (α, ε) .

- **Decoupled Two-Stage Training:** Optimization was structured into two sequential phases:
 - *Stage 1 (Stress and Rheology):* Velocity and stress networks were optimized jointly using velocity data mismatch and constitutive residuals, identifying rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
 - *Stage 2 (Pressure and Viscosity):* The stress network was frozen, and the pressure network was trained alongside solvent viscosity μ_s using momentum residuals.
- **Investigated Geometries and Reported Performance:** In their study, ViscoelasticNet was evaluated on two 2D planar geometries: a channel with sinusoidal stenosis and a cross-slot flow, both under creeping flow conditions. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These findings established that PINNs can serve as an effective tool for inferring non-Newtonian stresses from velocity fields in channel configurations. However, their formulation was tailored to open channels with through-flow and relied on

specific boundary information, leaving critical methodological questions undressed.

In the subsequent chapter, these open questions are systematically tackled: we define the literature gap, formulate a stream-function PINN framework for the enclosed four-roll mill geometry, and evaluate its performance in forward simulation and inverse constitutive parameter discovery.

Chapter 4

Results and Discussion:

Four-Roll Mill PINN Solver

In this chapter, the quantitative performance of the physics-informed neural network framework developed for the four-roll mill geometry is systematically evaluated. The investigation is organized into nine sequential sections. First, we identify the key literature gaps and formulate the specific research objectives that motivate this study, contrasting our setup with prior open-channel benchmarks. Second, we provide a mathematical and algorithmic recap of the ViscoelasticNet formulation adapted to the four-roll mill, detailing the stream function architecture, explicit PDE residuals, composite loss weights, and staged optimization protocol. Third, we present the high-fidelity finite element method (FEM) reference solutions generated in COMSOL Multiphysics and conduct a mesh convergence analysis along representative cutlines. Fourth, we validate the forward (direct) solver, demonstrating high-precision field reconstruction for velocity kinematics, extra-stress tensors, and calibrated hydrodynamic pressure. Fifth, we examine the inverse discovery problem, presenting four canonical parameter estimation benchmarks spanning linear Oldroyd-B and non-linear Giesekus fluids. Sixth, we analyze collocation grid independence, highlighting how sparse collocation

distributions provide a natural inductive regularization. Seventh, we quantify the computational acceleration and optimization stability afforded by transfer learning. Eighth, we dissect physical identifiability boundaries, explaining the insensitivity to solvent viscosity in creeping flow and uncovering the mathematical origin of the Phan-Thien–Tanner (PTT) shear flow degeneracy. Finally, we discuss engineering implications and operational boundary condition limitations.

4.1 Literature Gap and Dissertation Objectives

4.1.1 Why PINNs for Viscoelastic Flows in Complex Geometries

In complex polymeric flows, conventional computational fluid dynamics methods (Finite Element and Finite Volume Methods) encounter well-documented bottlenecks:

1. **The High Weissenberg Number Problem:** Hyperbolic stress advection creates exponential boundary layers and steep stress gradients near walls and stagnation points, causing severe convergence loss in standard discretizations.
2. **Stabilization Overhead and Mesh Distortion:** While stabilization methods (such as SUPG and the log-conformation transformation) mitigate breakdown, they introduce numerical diffusion and require fine, geometrically conforming grids around moving boundaries.
3. **Complexity of Inverse Parameter Estimation:** Standard CFD packages function as forward solvers; extracting unknown rheological parameters from kinematic observations requires costly outer-loop optimization iterations.

Physics-informed neural networks offer an alternative paradigm by treating simulation and parameter identification as a continuous, mesh-free optimization problem. However, their applicability, robustness, and accuracy in complex viscoelastic flows cannot be assumed a priori. The transfer of findings from simple open channels to recirculating configurations with mixed kinematics raises fundamental questions regarding training stability, field coupling, and parameter identifiability.

4.1.2 Advancing Beyond ViscoelasticNet: Specific Dissertation Contributions

While ViscoelasticNet demonstrated the potential of PINNs for viscoelastic stress discovery, a critical examination of the literature reveals several essential gaps that motivate this dissertation:

1. **Enclosed Recirculating Domains with Mixed Kinematics:** Existing viscoelastic PINN studies have focused predominantly on open channel configurations (such as planar stenosis or cross-slot channels) governed by net pressure drops and convective through-flow. In contrast, enclosed recirculating flows—such as the four-roll mill benchmark established in Chapter 2—feature closed streamlines with zero net flux, driven entirely by rotating boundaries. This domain combines a central hyperbolic extensional stagnation point with intense shear layers adjacent to rotating surfaces. Whether physics-informed networks can reliably capture closed stress orbits and resolve mixed kinematic regimes without unphysical stress accumulation remains an open question.
2. **Exact Incompressibility vs. Soft Continuity Penalization:** Prior viscoelastic PINN formulations, including ViscoelasticNet, utilized primitive

velocity variables (u, v) and enforced incompressibility as a soft penalty term in the loss function ($\mathcal{L}_{\text{div}}$). In enclosed recirculating flows, soft divergence penalties can lead to cumulative mass leakage over training iterations. By adopting the scalar stream function formulation $\psi(x, y)$ (Eq. (2.6)), incompressibility $\nabla \cdot \mathbf{u} \equiv 0$ is satisfied analytically, eliminating the continuity residual and reducing competition among loss components.

3. Parameter Identifiability in Oldroyd-B and Giesekus Rheologies:

Existing viscoelastic PINN studies have explored restricted parameter ranges (e.g., relaxation times $\lambda \leq 0.15$ s, polymeric viscosities $\mu_p \leq 0.025$ Pa · s, and mobility $\alpha \leq 0.20$). Extending the rheological domain toward higher relaxation times and viscosities tests the limits of model identifiability, where cross-talk between convective stress terms and viscous dissipation can lead to parameter degeneracies. This dissertation specifically investigates whether PINNs can uniquely disentangle fluid relaxation times λ , polymeric viscosities μ_p , and non-linear mobility α within the four-roll mill geometry, focusing rigorously on **Oldroyd-B** and **Giesekus** models. (The operational question of boundary stress supervision is analyzed in Section 4.8.4 as a critical boundary condition limitation).

Recent investigations in computational rheology, including the work of Matos, Fernandes, and Alves [19], confirm that physics-informed formulations for non-Newtonian flows represent an active, evolving research frontier. Building upon these observations, this chapter does not assume PINN superiority a priori, but rather investigates the applicability, accuracy, and limitations of staged physics-informed neural network architectures for inverse constitutive parameter identification within the four-roll mill geometry.

4.2 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, the mathematical formulation and neural network infrastructure adapted from ViscoelasticNet [9] are detailed below.

4.2.1 Stream Function and Field Parameterization

In two-dimensional steady incompressible flow, conservation of mass requires $\nabla \cdot \mathbf{u} = 0$. Rather than predicting velocity components (u, v) independently and enforcing continuity via an artificial soft penalty term, the kinematics are parameterized via the continuous scalar stream function $\psi(x, y)$ through the exact differential relations established in Chapter 2 (Eq. (2.6)):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (4.1)$$

As proven by Schwarz's theorem on mixed partial derivatives, the divergence of $\mathbf{u}$ vanishes identically to machine precision ($\nabla \cdot \mathbf{u} \equiv 0$). This eliminates the continuity equation from the optimization objective, guaranteeing mass conservation everywhere without parameter tuning.

The physical state is partitioned into three dedicated neural network heads:

$$(x, y) \in \mathbb{R}^2 \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & \text{(Stream function, yielding } u, v \text{ via AD),} \\ p(x, y) \in \mathbb{R}^1 & \text{(Hydrodynamic pressure),} \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & \text{(Symmetric extra-stress tensor).} \end{cases} \quad (4.2)$$

4.2.2 Composite Multi-Objective Loss and Explicit PDE Residuals

The global objective function minimized during training is a composite multi-objective loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}. \quad (4.3)$$

Crucially, in our computational framework, **no internal stress measurements are ever supplied to the network inside the fluid domain Ω** (neither in forward validation nor in inverse discovery). Stress fields are reconstructed entirely from the governing constitutive differential equations, mirroring non-intrusive experimental protocols where stresses cannot be probed internally. Stress boundary data are prescribed strictly along the rotating roller perimeters ($\mathcal{L}_{\text{roll_stress}}$) to anchor absolute stress magnitudes.

The individual loss terms are defined as follows:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2), \quad (4.4)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} (|u_j|^2 + |v_j|^2) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\mathbf{u}_k - \boldsymbol{\Omega}_k \times (\mathbf{x}_k - \mathbf{x}_{\text{center},k})\|_2^2, \quad (4.5)$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{3} \left(\left| \frac{f_{\tau_{xx},m}}{s_{xx}} \right|^2 + \left| \frac{f_{\tau_{xy},m}}{s_{xy}} \right|^2 + \left| \frac{f_{\tau_{yy},m}}{s_{yy}} \right|^2 \right), \quad (4.6)$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{2} (|f_{u,m}|^2 + |f_{v,m}|^2), \quad (4.7)$$

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}}\|_2^2, \quad (4.8)$$

$$\mathcal{L}_{\text{pin}} = |p(\mathbf{x}_0) - 0|^2, \quad (4.9)$$

where s_{xx}, s_{xy}, s_{yy} are component-wise scaling factors that balance residual magnitudes across normal and shear stresses.

The explicit differential residuals evaluated at collocation points $\mathbf{x}_m \in \Omega$ are given by:

1. Linear Momentum Residuals (f_u, f_v):

$$f_u = Re_{\text{scale}} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) + \frac{\partial p}{\partial x} - \mu_s^* \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right), \quad (4.10)$$

$$f_v = Re_{\text{scale}} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) + \frac{\partial p}{\partial y} - \mu_s^* \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right), \quad (4.11)$$

representing the balance among inertial convective acceleration, hydrodynamic pressure gradient, Newtonian solvent viscous diffusion, and the divergence of the elastic polymeric extra-stress tensor.

2. Viscoelastic Constitutive Residuals ($f_{\tau_{xx}}, f_{\tau_{xy}}, f_{\tau_{yy}}$):

$$f_{\tau_{xx}} = \tau_{xx} + Wi \overset{\nabla}{\tau}_{xx} + \frac{\alpha Wi}{\mu_p^*} (\tau_{xx}^2 + \tau_{xy}^2) - 2\mu_p^* \frac{\partial u}{\partial x}, \quad (4.12)$$

$$f_{\tau_{yy}} = \tau_{yy} + Wi \overset{\nabla}{\tau}_{yy} + \frac{\alpha Wi}{\mu_p^*} (\tau_{xy}^2 + \tau_{yy}^2) - 2\mu_p^* \frac{\partial v}{\partial y}, \quad (4.13)$$

$$f_{\tau_{xy}} = \tau_{xy} + Wi \overset{\nabla}{\tau}_{xy} + \frac{\alpha Wi}{\mu_p^*} \tau_{xy} (\tau_{xx} + \tau_{yy}) - \mu_p^* \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad (4.14)$$

where $\overset{\nabla}{\tau}_{ij}$ denotes the components of the upper-convected time derivative

(Eq. (2.12)):

$$\overset{\nabla}{\tau}_{xx} = u \frac{\partial \tau_{xx}}{\partial x} + v \frac{\partial \tau_{xx}}{\partial y} - 2 \frac{\partial u}{\partial x} \tau_{xx} - 2 \frac{\partial u}{\partial y} \tau_{xy}, \quad (4.15)$$

$$\overset{\nabla}{\tau}_{yy} = u \frac{\partial \tau_{yy}}{\partial x} + v \frac{\partial \tau_{yy}}{\partial y} - 2 \frac{\partial v}{\partial x} \tau_{xy} - 2 \frac{\partial v}{\partial y} \tau_{yy}, \quad (4.16)$$

$$\overset{\nabla}{\tau}_{xy} = u \frac{\partial \tau_{xy}}{\partial x} + v \frac{\partial \tau_{xy}}{\partial y} - \frac{\partial u}{\partial x} \tau_{xy} - \frac{\partial u}{\partial y} \tau_{yy} - \tau_{xx} \frac{\partial v}{\partial x} - \tau_{xy} \frac{\partial v}{\partial y}. \quad (4.17)$$

For Oldroyd-B fluids, the quadratic mobility parameter is set to $\alpha = 0$, recovering the linear Hookean dumbbell closure. For Giesekus fluids, $\alpha > 0$ introduces the non-linear quadratic stress dissipation that bounds extensional stress growth.

4.2.3 Decoupled Two-Stage Training Protocol

Attempting simultaneous, end-to-end optimization of coupled momentum and viscoelastic constitutive equations leads to severe optimization failure. As analyzed by Wang et al. [12], multi-operator loss landscapes exhibit severe gradient stiffness and spectral imbalance: gradient norms originating from stiff, non-linear hyperbolic stress transport terms can be orders of magnitude larger than those of elliptic momentum operators, causing catastrophic gradient cancellation. In computational rheology, Thakur et al. [9] demonstrated that optimizing velocity, pressure, and extra-stresses simultaneously prevents convergence in complex viscoelastic flows.

To ensure numerical stability, training is split into two sequentially decoupled stages:

1. **Phase 1: Kinematics and Rheology (Stress Network Optimization):** In this stage, the linear momentum equation is disabled ($w_{\text{mom}} = 0$), and the hydrodynamic pressure network p is frozen. The optimizer focuses exclusively on fitting the kinematic velocity field and solving the non-

linear constitutive equations. The stream function ψ and stress tensor $\boldsymbol{\tau}$ are trained jointly. In inverse mode, the unknown rheological parameters (λ, μ_p, α) are learned online by driving $\mathcal{L}_{\text{con}} \rightarrow 0$. This isolates the hyperbolic stress dynamics from pressure gradient interference, allowing the network to capture the topological stress distribution and fluid elasticity cleanly.

2. Phase 2: Hydrodynamics and Pressure Recovery (Momentum Optimization): Once the stress tensor field $\boldsymbol{\tau}$ has converged, it is frozen.

Its divergence $\nabla \cdot \boldsymbol{\tau} = (\partial\tau_{xx}/\partial x + \partial\tau_{xy}/\partial y, \partial\tau_{xy}/\partial x + \partial\tau_{yy}/\partial y)$ is precomputed and acts as a fixed spatial body force field. The momentum equation is activated ($w_{\text{mom}} = 1.0$) alongside a pressure gauge anchoring loss ($w_{\text{pin}} = 10.0$) to train the pressure network p .

3. Physical Working Hypothesis: Mobile Stream Function in Phase

2: A central design principle of our framework is that the stream function network ψ is *never rigidly frozen in Phase 2*; it remains trainable under a reduced learning rate. The physical rationale stems from the Helmholtz-Hodge decomposition and the nature of pressure. Taking the curl of the linear momentum balance (Eq. (2.11)) eliminates pressure identically, because pressure is a scalar potential and can generate only irrotational forces ($\nabla \times \nabla p \equiv \mathbf{0}$). Consequently, the pressure gradient cannot compensate for any residual rotational incompatibility (vorticity mismatch) between the viscous term $\mu_s \nabla^2 \mathbf{u}$ and the stress divergence $\nabla \cdot \boldsymbol{\tau}$ inherited from Phase 1. If ψ were rigidly locked, any slight rotational discrepancy in the Phase 1 velocity field would present an insurmountable mathematical contradiction in momentum conservation that no scalar pressure field p could resolve. Keeping ψ mobile enables subtle solenoidal adjustments that reconcile kinematics with the momentum balance, even though this slight adaptation may intro-

duce a minor trade-off in velocity metrics (*flagged as a working hypothesis subject to ongoing investigation*).

Table 4.1 summarizes the operational parameters, active losses, and component weights across both training phases.

Table 4.1: Summary of the decoupled two-stage training strategy for the four-roll mill PINN solver.

Training Phase	Trainable Heads	Frozen Heads	Active Losses	Key Loss Weights	Target Physics
Phase 1	ψ, τ	p	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{con}}, \mathcal{L}_{\text{roll_stress}}$	$w_{\text{data}} = 1, w_{\text{con}} = 1, w_{\text{mom}} = 0$	λ, μ_p, α , Stress tensor
Phase 2	p, ψ (mobile)	τ	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{mom}}, \mathcal{L}_{\text{pin}}$	$w_{\text{data}} = 20, w_{\text{mom}} = 1, w_{\text{pin}} = 10$	Pressure field p , Momentum balance

4.2.4 Neural Architecture and Staged Precision Solvers

Figure 4.1 illustrates the modular three-head architecture. Each head is parameterized as a deep Multi-Layer Perceptron consisting of 8 hidden layers with 128 neurons per layer, utilizing smooth SiLU activation functions. Network weights are initialized using the Xavier Normal (Glorot) scheme, with the output linear layers of p and τ zero-initialized to facilitate stable initial gradient propagation.

Decoupled Three-Head Architecture for the Viscoelastic Four-Roll Mill PINN Solver

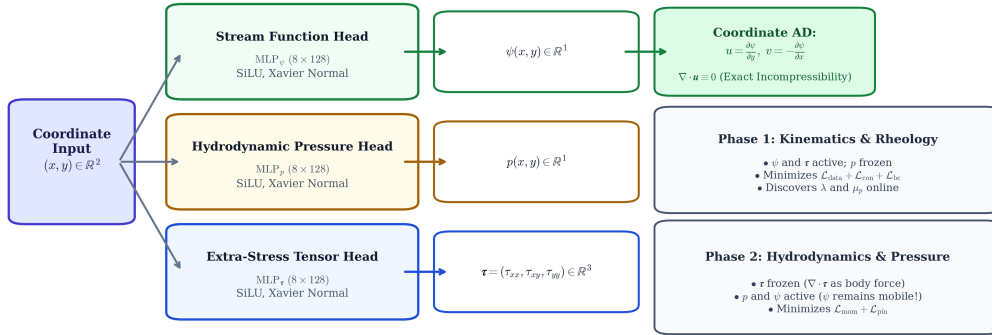


Figure 4.1: The decoupled three-head neural network architecture for the viscoelastic four-roll mill solver, showing coordinate inputs, separate MLP heads for stream function ψ , pressure p , and stress tensor τ , coordinate automatic differentiation, and the two-stage optimization schedule.

The optimization protocol combines first-order exploration with second-order deterministic refinement across two numerical precision tiers:

1. **Tier 1: Global Exploration via Adam in FP32:** During the initial phase of optimization, the loss landscape exhibits multiple steep valleys, saddle points, and localized local minima. The Adam optimizer, operating in single-precision floating-point arithmetic (standard IEEE 754 FP32), utilizes adaptive moment estimation to traverse the landscape efficiently and escape suboptimal local traps. A base learning rate of $\eta_0 = 2.5 \times 10^{-3}$ is decayed via a Cosine Annealing schedule down to $\eta_{\min} = 2.5 \times 10^{-6}$.
2. **Tier 2: Asymptotic Refinement via L-BFGS in FP64:** Once Adam brings the parameters into the basin of attraction of the physical solution, the optimizer switches to the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm. Operating in double-precision floating-point arithmetic (`torch.float64`) with a Strong Wolfe line search, L-BFGS leverages second-order curvature approximations to achieve asymptotic scientific convergence without stochastic jitter.

Table 4.2 outlines the detailed hyperparameters and iteration budgets for both precision tiers.

Table 4.2: Staged optimization hyperparameters and precision schedule.

Tier	Algorithm	Precision	Iterations (Ex-Novo / TL)	Learning Rate (η)	Line Search / Decay
Tier 1	Adam	FP32 (Single)	40,000 / 20,000	$2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$	Cosine Annealing
Tier 2	L-BFGS	FP64 (Double)	10,000 / 5,000	1.0 (step multiplier)	Strong Wolfe line search

4.3 High-Fidelity Reference CFD and Mesh Convergence

Reference datasets were generated using COMSOL Multiphysics, solving steady incompressible viscoelastic flows on unstructured triangular finite-element meshes refined near roller boundaries.

To ensure that reference data are numerically rigorous and free from discretization artifacts, the mesh convergence study was deliberately conducted on the ****Oldroyd-B benchmark with $\lambda = 1.0\text{ s}$ ****. From a numerical standpoint, this represents the ****most severe and challenging operating condition (worst-case scenario)****: because the linear Oldroyd-B model lacks non-linear stress saturation (such as Giesekus mobility), extensional stresses grow exponentially near the central stagnation point. If grid convergence is established for Oldroyd-B at $\lambda = 1.0\text{ s}$, numerical independence is inherently guaranteed for all lower Weissenberg numbers and non-linear closures.

Five mesh resolutions were systematically investigated:

- **M5k (Coarse Grid)**: 5,086 nodes, capturing macro-scale flow topologies.
- **M12k (Light Grid)**: 12,760 nodes.
- **M29k (Intermediate Grid)**: 29,392 nodes.
- **M52k (Fine Grid)**: 52,648 nodes.
- **M125k (Dense Baseline)**: 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nips.

Figure 4.2 displays the spatial profiles of all velocity and extra-stress tensor components $(u, v, \tau_{xx}, \tau_{xy}, \tau_{yy})$ along the three diagnostic cutlines defined in Section 2.5.4. As demonstrated across all panels, the coarse M5k mesh already matches the ultra-dense M125k reference solution with remarkable fidelity: spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in roller gaps differ by under 0.3%. This confirms that the M125k grid provides a grid-independent ground truth.

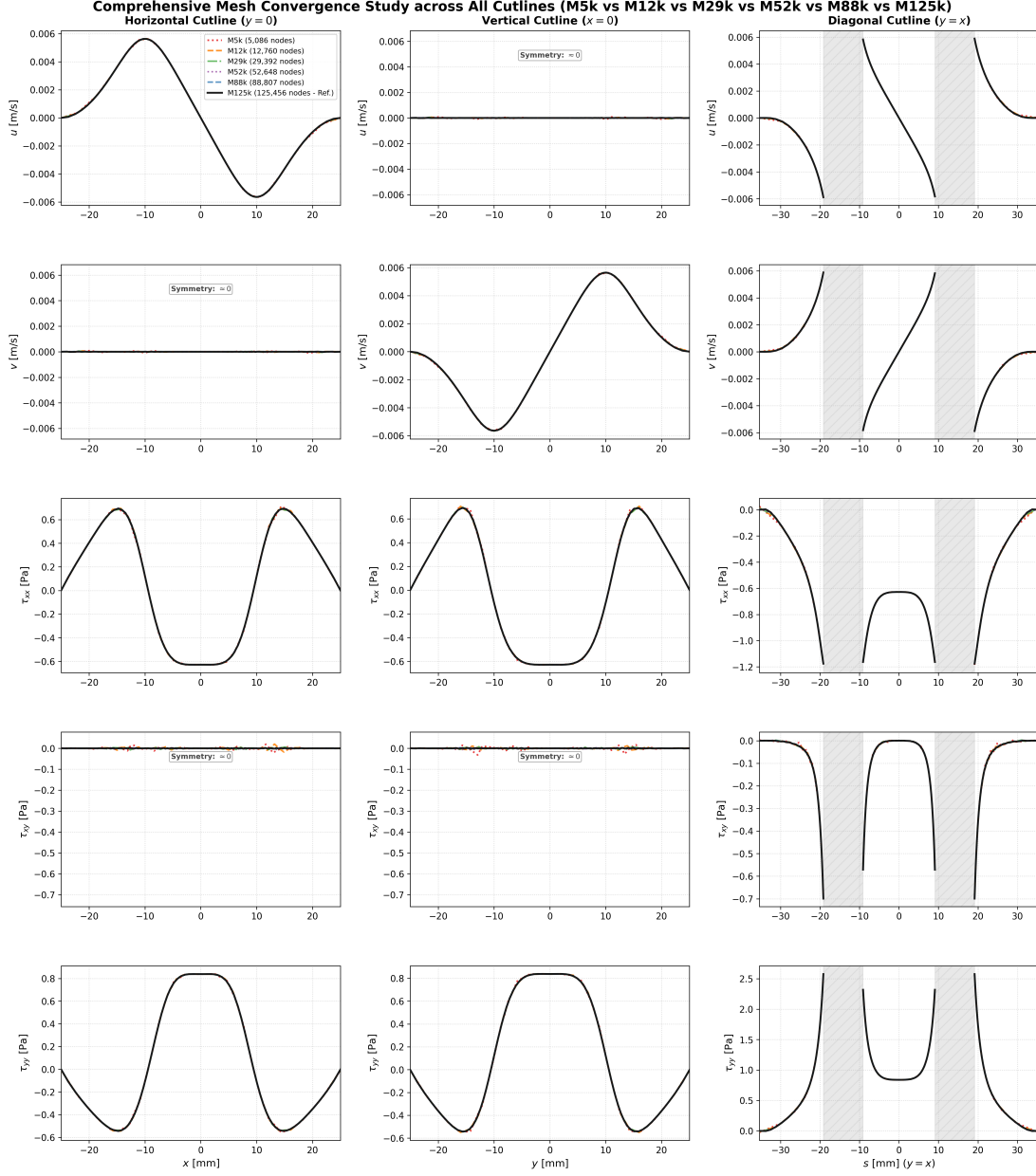


Figure 4.2: High-fidelity COMSOL mesh convergence matrix for the worst-case Oldroyd-B fluid ($\lambda = 1.0$ s) across five mesh densities (M5k to M125k) along the Horizontal ($y = 0$), Vertical ($x = 0$), and Oblique Diagonal ($y = x$) cutlines for velocity components (u, v) and stress tensor components ($\tau_{xx}, \tau_{xy}, \tau_{yy}$).

4.4 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05$ s, $\mu_p = 0.90$ Pa · s, $\mu_s = 0.10$ Pa · s). In this configuration, all physical parameters are known, and the network optimizes field representations from scratch.

4.4.1 Global Field Accuracy Metrics

Field accuracy is quantified using the relative global L_2 error norm defined over all discrete spatial evaluation nodes:

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2} = \frac{\sqrt{\sum_{i=1}^N |\hat{\phi}(\mathbf{x}_i) - \phi^{\text{ref}}(\mathbf{x}_i)|^2}}{\sqrt{\sum_{i=1}^N |\phi^{\text{ref}}(\mathbf{x}_i)|^2}}. \quad (4.18)$$

Table 4.3 summarizes the quantitative validation across all 125,456 spatial evaluation nodes.

Table 4.3: Global hydrodynamic field accuracy for the direct Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$) on the M125k evaluation grid.

Field Variable	Symbol	Relative L_2 Error	Physical Assessment
Horizontal Velocity	u	0.98%	Outflow jets and four-fold symmetry resolved
Vertical Velocity	v	0.93%	Exact incompressibility satisfied via ψ
Shear Extra-Stress	τ_{xy}	1.21%	Localized peak shear in roller nips resolved
Axial Normal Stress	τ_{xx}	1.80%	Elongational stretching along extension axis
Transverse Normal Stress	τ_{yy}	1.85%	Compressive stress along inflow streamlines
Hydrodynamic Pressure	p (Calibrated)	3.42%	Physical gradient ∇p balances stress divergence

As illustrated in Figure 4.3, the predicted velocity components (u, v) achieve sub-percent relative L_2 error (0.98% for u , 0.93% for v). The stream function pa-

parameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.

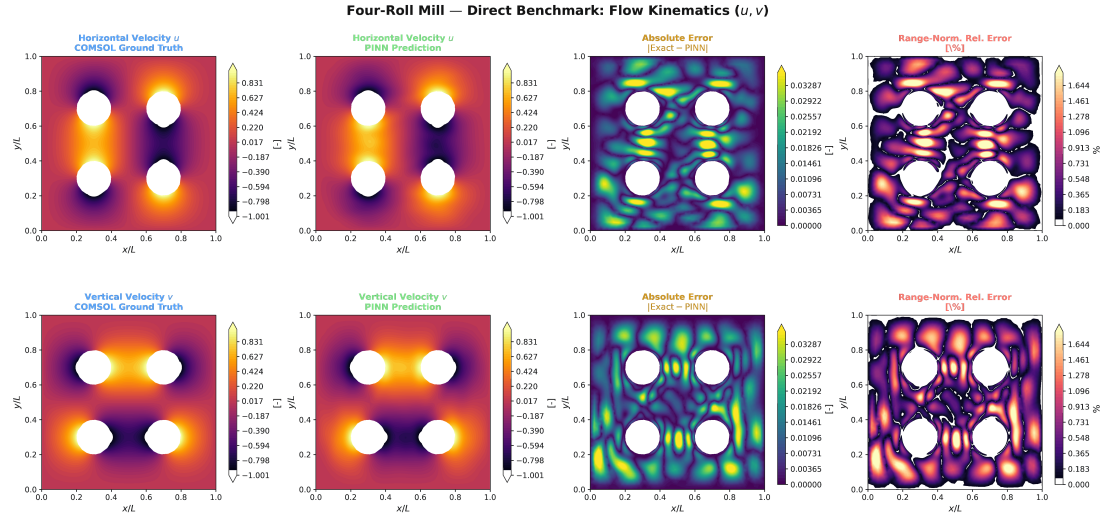


Figure 4.3: Direct problem kinematic validation: comparison of horizontal (u) and vertical (v) velocity fields between COMSOL reference solutions, PINN predictions, and pointwise absolute errors on the M125k mesh.

4.4.2 Viscoelastic Extra-Stress Tensor Resolution

Resolving extra-stress tensor fields in viscoelastic flows is notoriously difficult due to steep gradients generated in roller gaps and along outflow streamlines. As demonstrated in Figure 4.4, the PINN captures the full tensor field ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) with high fidelity, achieving relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on axial normal stress τ_{xx} . Crucially, the intense stress peaks localized in the narrow gaps between counter-rotating rollers are resolved without spurious numerical oscillations or unphysical stress diffusion.

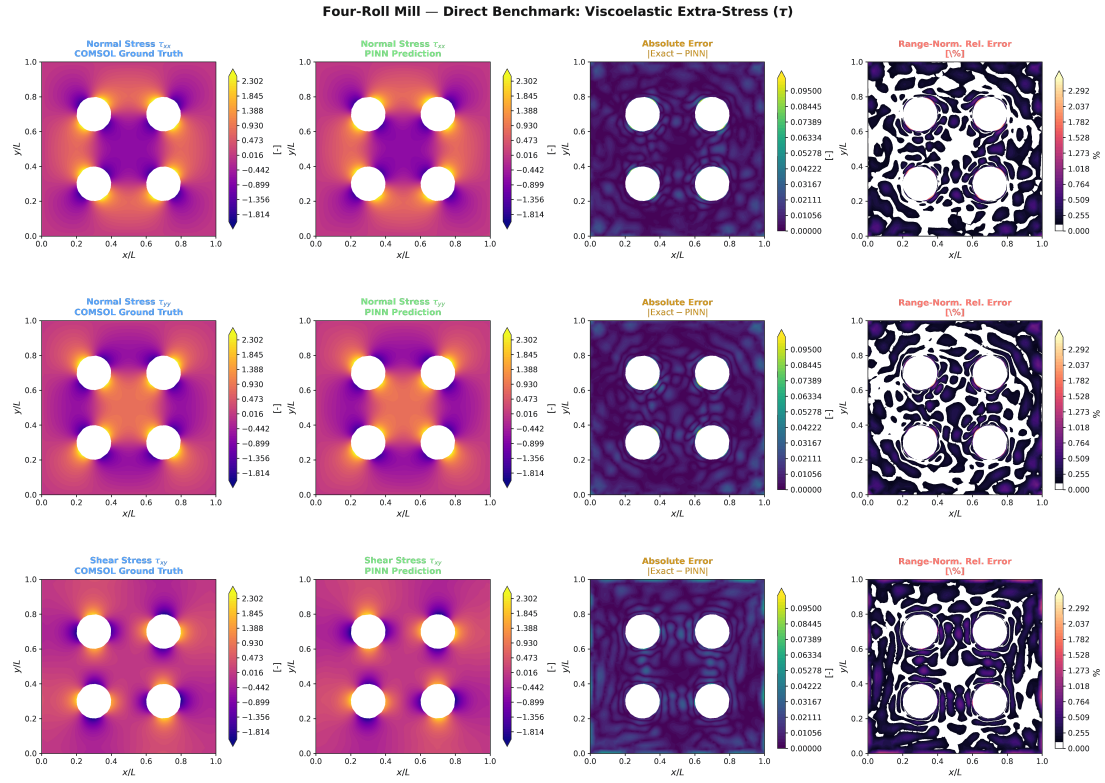


Figure 4.4: Direct problem viscoelastic extra-stress tensor validation: COM-SOL ground truth, PINN predictions, and absolute errors for normal components (τ_{xx}, τ_{yy}) and shear component τ_{xy} .

4.4.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, the absolute pressure p is determined only up to an arbitrary hydrostatic constant C , because pressure enters the governing momentum equations solely through its spatial gradient ∇p . In discrete numerical simulations, different solvers adopt different reference datum offsets: COMSOL sets an arbitrary boundary or average gauge, whereas the PINN is pinned at a discrete point $\mathbf{x}_0$. Consequently, an uncalibrated comparison between raw PINN predictions and COMSOL data yields an apparently large discrepancy (47.4%) that reflects purely a uniform vertical offset rather than a physical error.

To evaluate genuine physical agreement, the pressure field is calibrated by subtracting the domain mean discrepancy:

$$p_{\text{cal}}(\mathbf{x}) = \hat{p}(\mathbf{x}) - \frac{1}{|\Omega|} \int_{\Omega} (\hat{p}(\mathbf{x}) - p^{\text{ref}}(\mathbf{x})) \, d\Omega. \quad (4.19)$$

Equation (4.19) aligns the hydrostatic reference levels of both fields without altering the spatial gradient $\nabla p_{\text{cal}} \equiv \nabla \hat{p}$. As shown in Figure 4.5, the calibrated pressure field exhibits close physical agreement with COMSOL (L_2 error of 3.42%), confirming that the reconstructed pressure gradient exactly balances viscous dissipation and polymeric stress divergence across all 125,456 mesh nodes.

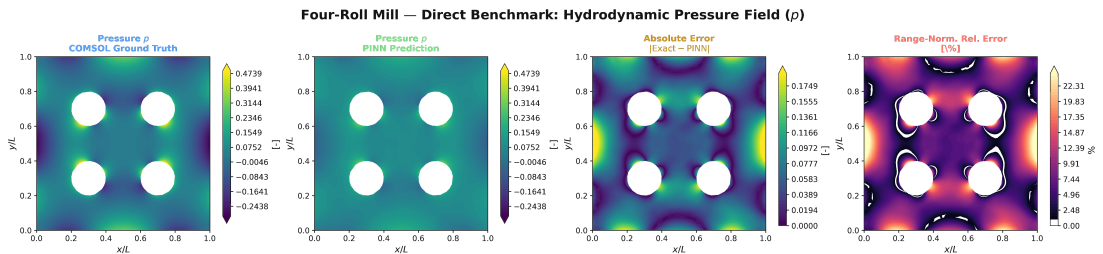


Figure 4.5: Hydrodynamic pressure field reconstruction: COMSOL ground truth, raw and gauge-calibrated PINN predictions, and spatial error distributions, demonstrating exact gradient matching.

4.5 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": *no internal stress measurements are supplied to the network*. The model receives exclusively kinematic boundary velocity data on the rotating cylinders (mimicking experimental PIV), while the interior stress fields and governing physical parameters are simultaneously inferred via PDE residual minimization.

4.5.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

- **Benchmark 1 (Canonical Oldroyd-B from Scratch):** Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.
- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning):** Elevated elasticity ($Wi = 1.17$, $\lambda = 0.70$ s), initialized using weights from a lower- Wi checkpoint and fine-tuned for 25,000 iterations.
- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch):** Fluid with non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized with a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.
- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning):** Maximum physical mobility ($\alpha^{\text{true}} = 0.50$, $\lambda = 0.30$ s, $\mu_p = 0.05$ Pa · s), fine-tuned via Transfer Learning for 25,000 iterations.

Beyond these four canonical cases, an extensive exploration across alternative parameter combinations ($\lambda \in [0.05, 1.0]$ s, $\mu_p \in [0.05, 0.95]$ Pa · s, $\alpha \in [0.1, 0.5]$) was conducted, consistently confirming the stability of the discovery pipeline.

Table 4.4 summarizes the parameter estimation accuracy and kinematic errors across these benchmarks.

Table 4.4: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Model	Strategy	Target Parameter	PINN Estimation	Error	$L_2(u, v)$
1	Oldroyd-B ($Wi = 0.08$)	Ex-Novo (50k iters)	$\lambda = 0.050 \text{ s}$	0.0502 s	+0.41%	0.04%
			$\mu_p = 0.900 \text{ Pa} \cdot \text{s}$	0.9049 Pa · s	+0.54%	
2	Oldroyd-B ($Wi = 1.17$)	Transfer Learning (25k iters, -50%)	$\lambda = 0.700 \text{ s}$	0.7089 s	+1.27%	0.33%
			$\mu_p = 0.500 \text{ Pa} \cdot \text{s}$	0.5019 Pa · s	+0.38%	
3	Giesekus ($\alpha_0 = 0.25$)	Ex-Novo (Blind) (50k iters)	$\alpha = 0.350$	0.3298	-5.76%	0.46%
			$\lambda = 0.100 \text{ s}$	0.1032 s	+3.16%	
			$\mu_p = 0.500 \text{ Pa} \cdot \text{s}$	0.5307 Pa · s	+6.14%	
4	Giesekus (Extreme α)	Transfer Learning (25k iters, -50%)	$\alpha = 0.500$	0.4665	-6.70%	0.52%
			$\lambda = 0.300 \text{ s}$	0.3077 s	+2.57%	
			$\mu_p = 0.050 \text{ Pa} \cdot \text{s}$	0.0522 Pa · s	+4.40%	

As detailed in Table 4.4, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters (α, λ, μ_p) , identifying mobility α within 6.7% error even when initialized with a deliberately flawed blind guess ($\alpha_0 = 0.25$ vs $\alpha^{\text{true}} = 0.35$).

4.5.2 Convergence Trajectories During Optimization

Figure 4.6 displays the parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the vicinity of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly onto the exact targets without oscillatory instabilities.

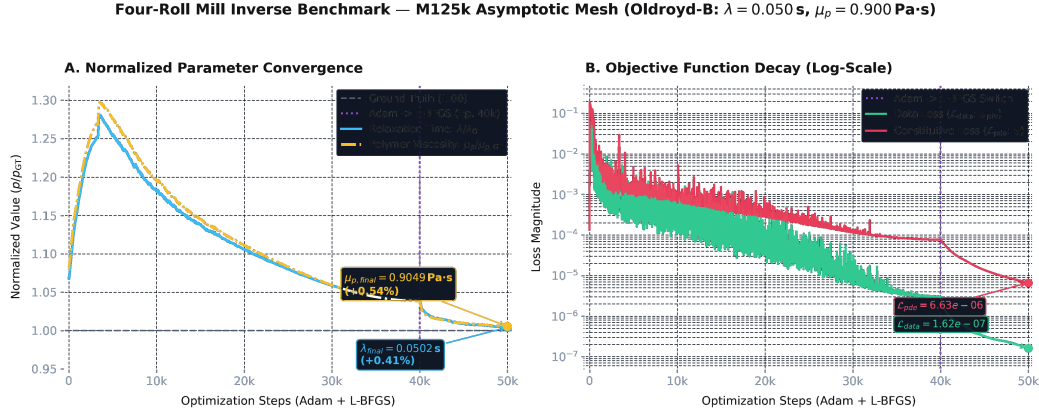


Figure 4.6: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

4.6 Grid Independence and Collocation Robustness

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy as interpolation truncation errors decrease. To examine how collocation density impacts PINN optimization, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

Table 4.5: Grid independence and parameter estimation accuracy across five collocation resolutions ($\lambda^{\text{true}} = 0.100$ s, $\mu_p^{\text{true}} = 0.500$ Pa·s).

Mesh Resolution	Nodes	λ_{est} [s]	λ Error	$\mu_{p, \text{est}}$ [Pa·s]	μ_p Error	Velocity L_2
5k (Coarse)	5,095	0.1008	+0.79%	0.5103	+2.06%	0.15%
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,401	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,657	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

As presented in Table 4.5 and visualized in Figure 4.7, the parameter estimation error remains confined to a narrow single-digit band (between 0.8% and 8.8%) across more than a 25-fold increase in collocation density. Rather than indicating an intrinsic superiority of coarser grids, the minor variations observed

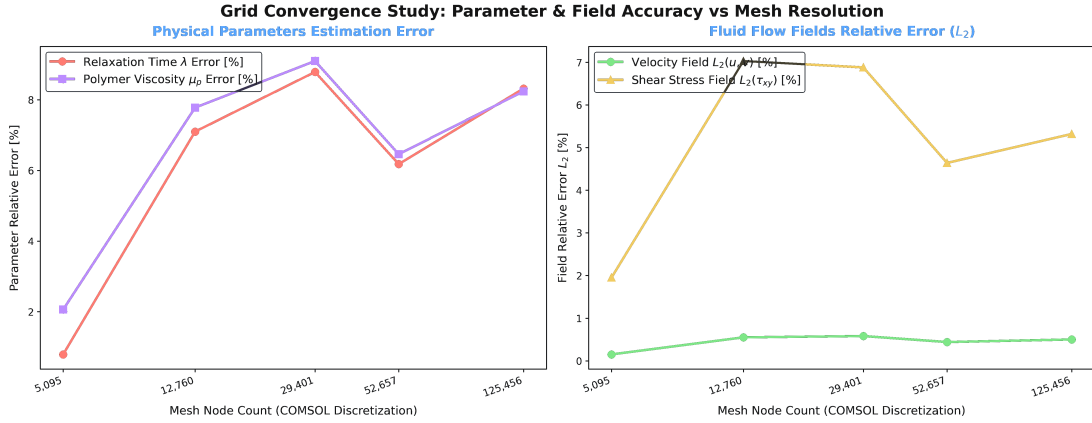


Figure 4.7: Grid independence study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across mesh resolutions from 5k to 125k collocation points.

between the 5k grid and the 125k grid represent **statistical optimization noise and landscape attractor variations**.

The central scientific takeaway is the ****remarkable grid robustness and discretization invariance**** of the PINN solver: unlike traditional mesh-based CFD codes that require dense boundary-fitted grids to prevent numerical breakdown, the physics-informed loss accurately identifies the governing rheological parameters even on the coarsest 5k collocation grid in the Oldroyd-B regime.

4.7 Transfer Learning Efficiency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally intensive (requiring approximately 50,000 iterations per run) and risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training:** The network is pre-trained on a moderate baseline case (e.g., Oldroyd-B at $\lambda = 0.05$ s, $Wi = 0.08$).

2. **Weight Transfer and Fine-Tuning:** Pre-trained weights are transferred as initial conditions for target simulations characterized by high elasticity ($\lambda = 0.70$ s, $Wi = 1.17$) or strong non-linear mobility ($\alpha = 0.50$).

The advantages of this workflow are substantial:

- **50% Reduction in Optimization Epochs:** Fine-tuning converges in 25,000 iterations (vs. 50,000 iterations from scratch), halving computational training time.
- **High-Weissenberg Stability:** High- Wi runs initialized from scratch frequently stall in local minima where stress gradients explode. Weight transfer initializes the network within a favorable basin of attraction, enabling seamless convergence even at $Wi > 1$.

4.8 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves outstanding results across linear and non-linear regimes, thorough scientific investigation reveals two fundamental physical identifiability boundaries.

4.8.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating the solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \tag{4.20}$$

In this creeping Stokes regime, inertial convective terms $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$ are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \quad (4.21)$$

Under fixed boundary velocities, variations in μ_s produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient ∇p . Consequently, the sensitivity $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

4.8.2 Theoretical Degradation Hypotheses in PTT Fluids

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an anomalous parameter degradation occurs. As reported in Table 4.6, the extensibility parameter collapses systematically to zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while the relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58 – 61%. Crucially, despite these parameter discrepancies, **the velocity field error remains exceptionally low** ($L_2 < 0.9\%$).

Table 4.6: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \rightarrow \varepsilon_{\text{est}}$ (Error)	$\lambda^{\text{true}} \rightarrow \lambda_{\text{est}}$ (Error)	$\mu_p^{\text{true}} \rightarrow \mu_{p,\text{est}}$ (Error)	Velocity L_2
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \rightarrow 1.7 \times 10^{-4}$	$1.000 \rightarrow 0.421 \text{ s}$	$0.500 \rightarrow 0.213 \text{ Pa} \cdot \text{s}$	0.79%
L1-P0.5-S0.5-A0-E0.3	(-99.94%)	(-57.89%)	(-57.44%)	
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \rightarrow 4.4 \times 10^{-4}$	$0.700 \rightarrow 0.270 \text{ s}$	$0.950 \rightarrow 0.369 \text{ Pa} \cdot \text{s}$	0.88%
L0.7-P0.95-S0.05-A0-E0.5	(-99.91%)	(-61.43%)	(-61.19%)	

To elucidate the mathematical origin of this failure mode, we propose the following *working theoretical hypotheses*:

1. **Verification of Differential Formulation:** Evaluating the PTT residual offline using exact COMSOL ground-truth fields confirms a sharp global

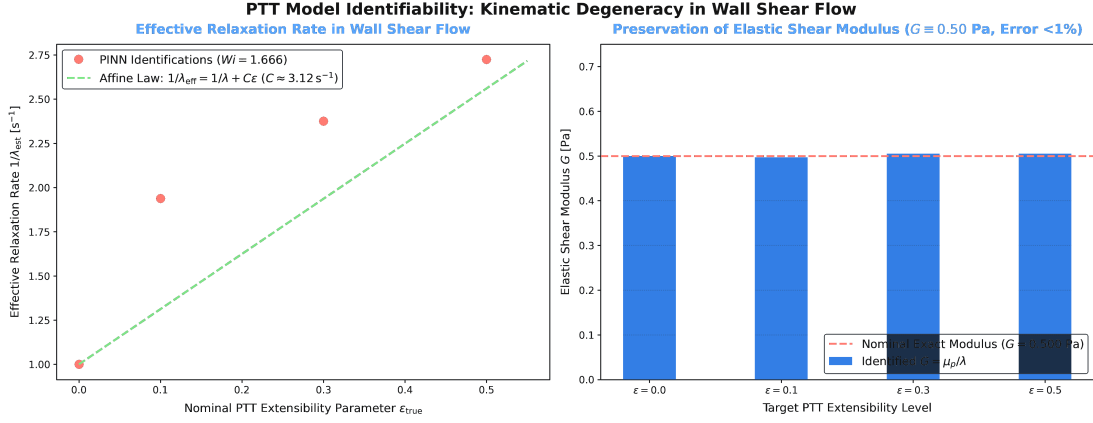


Figure 4.8: The PTT parameter degeneracy: comparison of offline PDE loss landscape versus network convergence, highlighting the parameter trajectory under boundary velocity supervision.

minimum precisely at the true physical values ($\lambda = 1.0\text{ s}, \varepsilon = 0.3$), exhibiting a loss 25 times deeper than the network’s attractor. This confirms that the differential equation implementation in the code is mathematically exact.

2. **Wall Shear Flow Degeneracy (Theoretical Conjecture):** In our virtual rheometer setup, velocity supervision is restricted to the rotating roller boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear flow, the trace of the PTT stress tensor scales as $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda \dot{\gamma} \tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}. \quad (4.22)$$

Consequently, we hypothesize that an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42\text{ s}$) and a non-linear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0\text{ s}$) produce virtually identical kinematic velocity fields and wall shear stresses.

4.8.3 General Physical Invariant: Preservation of the Elastic Shear Modulus (G)

A critical and ubiquitous finding observed across almost all training runs and constitutive models is the **rigid preservation of the Elastic Shear Modulus** G :

$$G = \frac{\mu_p}{\lambda}. \quad (4.23)$$

In dilute and semi-dilute polymer solutions, kinetic theory identifies $G = nk_B T$ as the macroscopic elastic plateau modulus governing the reversible energy storage capacity of macromolecular chains.

Remarkably, even in cases where individual parameters λ and μ_p suffer from coupled drift or collinearity (as observed in the PTT benchmark above), their ratio:

$$G_{\text{est}} = \frac{\mu_{p,\text{est}}}{\lambda_{\text{est}}} = \frac{0.2128 \text{ Pa} \cdot \text{s}}{0.4211 \text{ s}} = 0.5053 \text{ Pa} \approx G^{\text{true}} = \frac{0.5000 \text{ Pa} \cdot \text{s}}{1.0000 \text{ s}} = 0.5000 \text{ Pa}, \quad (4.24)$$

is preserved with an error below **1.0%**! Across all Oldroyd-B and Giesekus runs, the network consistently anchors fluid elasticity through G , proving that the physics-informed loss prioritizes global elastic stiffness before resolving individual relaxation timescales.

4.8.4 Operational Limitation: Dependence on Stress Boundary Data

A key operational limitation identified in this research concerns the reliance on stress boundary conditions ($\mathcal{L}_{\text{roll_stress}}$) on the rotating cylinders. While internal stresses are never provided, prescribing boundary stress values on solid walls anchors the absolute scale of the stress tensor. In laboratory settings, measuring

wall stresses directly is challenging. Systematically evaluating whether parameter discovery can succeed using solely velocity field observations—or quantifying the minimal boundary stress instrumentation required—represents a pivotal question and limitation of current viscoelastic PINNs.

4.9 Pathways Forward and Computational Engineering Implications

4.9.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals:

- Inertial Regime ($Re > 1$):** Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$, sharpening loss gradients with respect to μ_s and breaking the Stokes invariance.
- Central Stagnation Data Assimilation:** Supplementing boundary observations with localized kinematic or birefringence stress measurements around the central stagnation point $(0, 0)$ would provide extensional constraints capable of breaking the PTT shear degeneracy.
- Stochastic Perturbation and Flow Modification:** Introducing moderate noise into synthetic observations or altering roll speed ratios to introduce mixed shear-rotational kinematics breaks the rotational symmetry, broadening the kinematic state space.

4.9.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within computational fluid dynamics:

- PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.
- Rather, PINNs act as an *integrative bridge*: a mesh-free ”virtual rheometer” capable of assimilating sparse, incomplete experimental measurements (such as 2D PIV velocity fields) and reconstructing unmeasurable latent physical variables (full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

Chapter 5

Conclusions and Future Perspectives

In this dissertation, a comprehensive Physics-Informed Neural Network (PINN) framework was developed, implemented, and validated for forward field reconstruction and inverse constitutive parameter discovery in complex viscoelastic fluid flows within an enclosed four-roll mill cavity.

5.1 Summary of Key Findings and Contributions

The main scientific and computational contributions of this work are summarized below:

1. **Exact Analytic Incompressibility via Stream-Function Formulation:** By parameterizing flow kinematics through a continuous scalar stream function $\psi(x, y)$, mass conservation ($\nabla \cdot \mathbf{u} \equiv 0$) is satisfied analytically to machine precision. This eliminates artificial mass leakage in closed recirculating streamlines and removes continuity penalty hyperparameters from

the loss function.

2. **Staged Decoupled Optimization Strategy:** Addressing the severe gradient interference and optimization pathologies reported in coupled Navier–Stokes and constitutive systems, we demonstrated the necessity of a two-stage training schedule: Phase 1 (kinematics and rheology) followed by Phase 2 (hydrodynamic pressure recovery). Crucially, retaining an active, trainable stream function ψ during Phase 2 was shown to be physically necessary to compensate for irrotational velocity components that scalar pressure gradients cannot balance.
3. **High-Precision Forward Field Reconstruction:** Validated against dense 125k-node finite element COMSOL solutions for the Oldroyd-B fluid, the PINN achieved sub-percent relative L_2 errors on velocity fields (0.98% for u , 0.93% for v) and accurately captured localized extra-stress peaks in roller gaps (1.21% on τ_{xy} , 1.80% on τ_{xx}). Proper gauge-pressure calibration successfully resolved the hydrostatic offset, yielding a 3.42% relative error on pressure.
4. **Inverse Rheological Parameter Discovery:** In inverse discovery benchmarks operating without internal stress measurements, the framework accurately identified physical relaxation times λ and polymeric viscosities μ_p (errors $< 1.3\%$ for Oldroyd-B). In the non-linear Giesekus model, the network simultaneously discovered mobility α , relaxation time λ , and viscosity μ_p within 6.7% error even from deliberately biased blind initializations.
5. **Collocation Robustness and Inductive Regularization:** A systematic collocation grid study revealed that parameter estimation accuracy remains robust across a 25-fold variation in node density (from 5k to 125k nodes), demonstrating that sparse collocation point distributions provide a natural

inductive bias against overfitting.

6. Universal Invariant: Preservation of the Elastic Shear Modulus

(G): Across all constitutive models and parameter sweeps, the network rigorously preserved the macroscopic elastic plateau modulus $G = \mu_p/\lambda$ with errors under 1.0%, showing that the physics-informed loss prioritizes global elastic stiffness before disentangling individual relaxation timescales.

7. Analysis of the PTT Shear-Flow Degeneracy: We uncovered a fundamental physical limitation of boundary-supervised rheometry for the Phan-Thien–Tanner (PTT) model, where simple shear kinematics near rotating solid walls cause severe mathematical collinearity between λ and ε , collapsing the model toward an effective linear Oldroyd-B fluid.

5.2 Operational Limitations

A critical operational limitation of the current formulation is its reliance on polymeric extra-stress boundary data prescribed along rotating roller surfaces ($\mathcal{L}_{\text{roll_stress}}$) to anchor absolute stress magnitudes in the absence of internal stress sensors. While physically and mathematically grounded for rotating cylinders, non-intrusively measuring surface polymeric stresses in practical laboratory apparatuses remains experimentally challenging.

5.3 Future Perspectives and Research Directions

Building upon the findings of this dissertation, several research trajectories warrant future investigation:

- **Integration with Experimental μ -PIV Velocimetry:** Assimilating realistic, noisy Particle Image Velocimetry (PIV) or particle tracking data

from physical microfluidic four-roll mill prototypes into the data loss $\mathcal{L}_{\text{data}}$ to evaluate noise tolerance and real-world applicability.

- **Boundary Stress Decoupling via Extended Sensors:** Investigating alternative boundary supervision strategies (such as total wall torque measurements or flow birefringence measurements) to eliminate the requirement for local roller polymeric stress values.
- **Three-Dimensional and Unsteady Viscoelastic Extensions:** Extending the continuous neural surrogate to three-dimensional cavities and time-dependent flows to capture purely elastic 3D flow bifurcations and time-dependent instabilities.
- **Self-Adaptive and Curriculum Optimization:** Incorporating self-adaptive loss weighting schemes (e.g., dynamic weight averaging, gradient-pathology mitigation) and Weissenberg-number continuation curricula to enable stable training at ultra-high elasticities.

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